ISO 9001, QS 9000, ISO 14000

PASSIVE SYSTEM ALLIANCE

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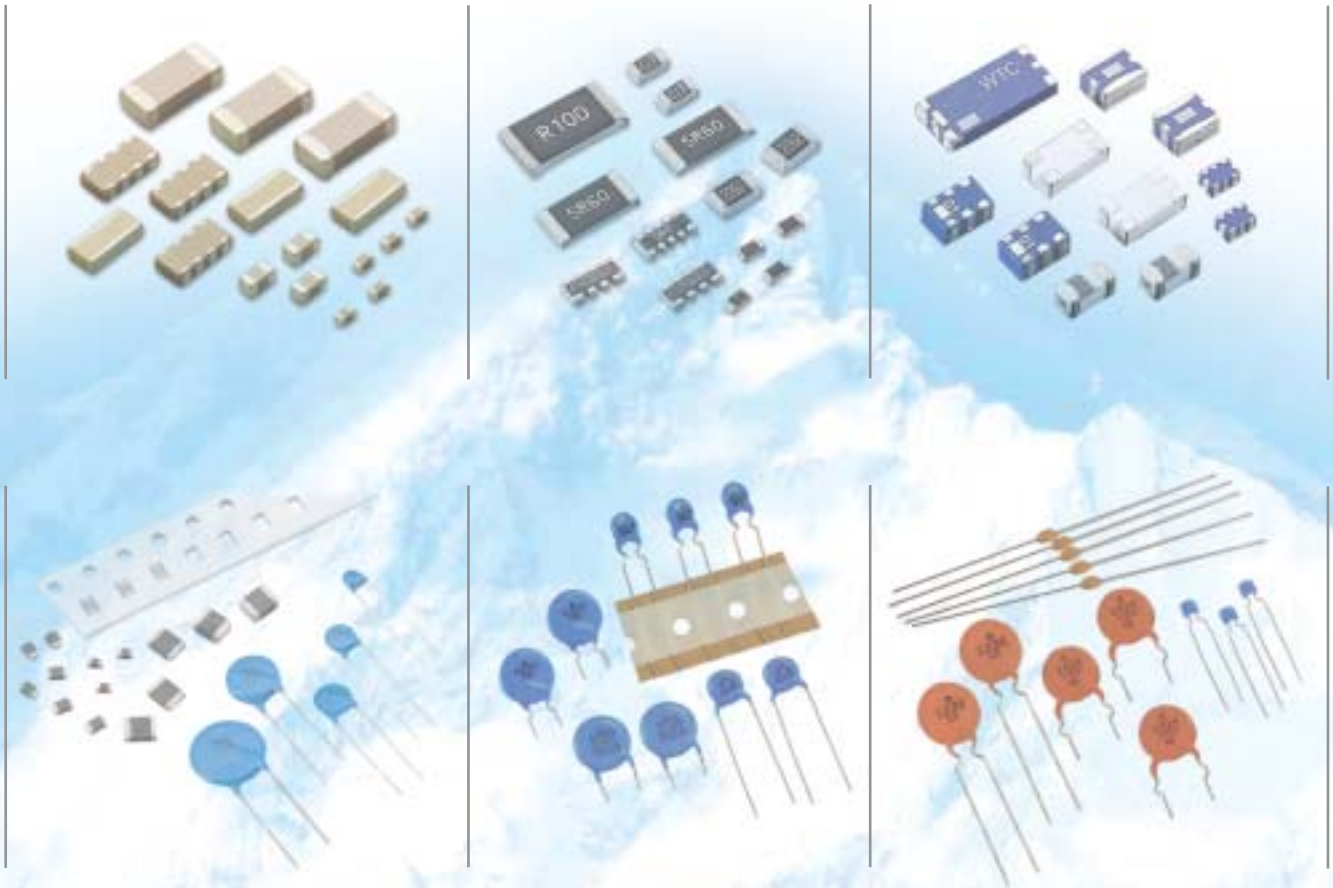
# Walsin Technology Corporation

## Multilayer Ceramic Capacitors

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# Table of Contents


| Subject | Page |
|---|---|
| Quick Product Information | 1 |
| The Outlines and External Dimensions of Capacitors | 2 |
| General Purpose Capacitors | 3 |
| High Capacitance Capacitors | 7 |
| Low Profile Capacitors | 9 |
| Ultra-small (0201) Capacitors | 10 |
| Middle and High Voltage Capacitors | 11 |
| Safety Certificated Capacitors (X1/Y2) | 16 |
| Safety Certificated Capacitors (X2/Y3) | 17 |
| High Q and Low ESR Capacitors | 19 |
| Microwave Capacitors | 21 |
| Open-mode Design Capacitors | 23 |
| Capacitor Arrays | 25 |
| Low Inductance Capacitors | 26 |
| **Appendix I :** Reliability Test Conditions and Requirements | 27 |
| **Appendix II :** General Information | 30 |

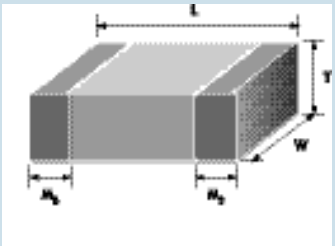
# Quick Product Information

## Quick Product Information

| Series | Dielectric | Size | Capacitance | Rated voltage | Page |
|---|---|---|---|---|---|
| General Purpose Caps | NP0 | 0402, 0603, 0805, 1206, 1210, 1812 | 0.5pF~0.039μF | 16V, 50V, 100V | 4 |
| | X7R | 0402, 0603, 0805, 1206, 1210, 1812 | 100pF~1μF | 10V, 16V, 25V, 50V, 100V | 5 |
| | Y5V | 0402, 0603, 0805, 1206, 1210, 1812 | 0.01μF~1μF | 10V, 16V, 25V, 50V, 100V | 6 |
| High Capacitance Caps | X7R | 0402, 0603, 0805, 1206, 1210, 1812 | 0.1μF~4.7μF | 10V, 16V, 25V, 50V | 8 |
| | X5R | 0402, 0603, 0805, 1206 | 0.027μF~10μF | 6.3V, 10V, 16V | 8 |
| | Y5V | 0402, 0603, 0805, 1206, 1210, 1812 | 0.15μF~47μF | 6.3V, 10V, 16V, 25V, 35V, 50V | 8 |
| Low Profile Caps | X5R | 0805, 1206, 1210 | 1μF~100μF | 6.3V, 10V | 9 |
| | Y5V | 0805, 1206, 1210 | 2.2μF~10μF | 10V, 16V | 9 |
| Ultra-small Caps | NP0 | 0201 | 0.5pF~100pF | 16V, 25V | 10 |
| | X7R | 0201 | 100pF~4700pF | 16V, 25V, 50V | 10 |
| | X5R | 0201 | 1000pF~0.022μF | 6.3V, 10V, 16V | 10 |
| Middle & High Voltage Caps | NP0 | 0603, 0805, 1206, 1210, 1808, 1812 | 0.5pF~6800pF | 200V, 250V, 500V, 630V, 1kV, 2kV, 3kV | 12 |
| | X7R | 0805, 1206, 1210, 1808, 1812 | 100pF~0.47μF | 200V, 250V, 500V, 630V, 1kV, 1.5kV, 2kV, 3kV | 14 |
| | Y5V | 0805, 1206, 1210, 1812 | 0.01μF~0.68μF | 200V, 250V | 15 |
| Safety Certificated Caps (X1/Y2) | NP0 | 1808, 1812 | 10pF~470pF | 250Vac | 16 |
| Safety Certificated Caps (X2/Y3) | NP0 | 1808, 1812 | 3.9pF~1000pF | 250Vac | 18 |
| | X7R | 1808, 1812 | 150pF~4700pF | 250Vac | 18 |
| High Q & Low ESR Caps | NP0 | 0402, 0603 | 0.5pF~3300pF | 16V, 50V, 100V | 20 |
| Microwave Caps | NP0 | 0402, 0603 | 0.1pF~22pF | 50V | 22 |
| Open-mode Design Caps | X7R | 0805, 1206, 1210, 1812 | 100pF~1μF | 100V, 200V, 250V, 500V | 24 |
| Capacitor Arrays | NP0 | 0612 (4x0603) | 10pF~470pF | 50V | 25 |
| | X7R | 0612 (4x0603) | 180pF~0.1μF | 16V, 50V | 25 |
| | Y5V | 0612 (4x0603) | 0.01μF~0.1μF | 50V | 25 |
| Low Inductance Caps | X7R | 0612 | 0.01μF~0.15μF | 50V | 26 |

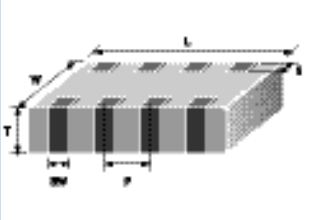
# The Outlines and External Dimensions of Capacitor

## Single Chip Capacitors

| Outline | Size Inch (mm) | L (mm) | W (mm) | T (mm)/Symbol | | Remark | $M_B$ (mm) |
|---|---|---|---|---|---|---|---|
|  | 0201 (0603) | 0.6±0.03 | 0.3±0.03 | 0.3±0.03 | L | # | 0.15±0.05 |
| | 0402 (1005) | 1.00±0.05 | 0.50±0.05 | 0.50±0.05 | N | # | 0.25 +0.05/-0.10 |
| | 0603 (1608) | 1.60±0.10 | 0.80±0.10 | 0.80±0.07 | S | | 0.40±0.15 |
| | | 1.60 +0.15/-0.10 | 0.80 +0.15/-0.10 | 0.80 +0.15/-0.10 | X | | |
| | 0805 (2012) | 2.00±0.15 | 1.25±0.10 | 0.60±0.10 | A | | 0.50±0.20 |
| | | | | 0.80±0.10 | B | | |
| | | | | 1.25±0.10 | D | # | |
| | | 2.00±0.20 | 1.25±0.20 | 1.25±0.20 | I | # | |
| | 1206 (3216) | 3.20±0.15 | 1.60±0.15 | 0.80±0.10 | B | | 0.60±0.20 |
| | | | | 0.95±0.10 | C | | |
| | | | | 1.25±0.10 | D | # | |
| | | 3.20±0.20 | 1.60±0.15 | 1.15±0.15 | J | # | |
| | | | 1.60±0.20 | 1.60±0.20 | G | # | |
| | | | 1.60+0.30/-0.10 | 1.60+0.30/-0.10 | P | # | |
| | 1210 (3225) | 3.20±0.30 | 2.50±0.20 | 0.95±0.10 | C | # | 0.75±0.25 |
| | | | | 1.25±0.10 | D | # | |
| | | 3.20±0.40 | 2.50±0.30 | 1.60±0.20 | G | # | |
| | | | | 2.00±0.20 | K | # | |
| | | | | 2.50±0.30 | M | # | |
| | 1808 (4520) | 4.50±0.40 | 2.03±0.25 | 1.25±0.10 | D | # | 0.75±0.25* |
| | | | | 2.00±0.20 | K | # | |
| | 1812 (4532) | 4.50±0.40 | 3.20±0.30 | 1.25±0.10 | D | # | 0.75±0.25* |
| | | | | 2.00±0.20 | K | # | |
| | | | | 2.50±0.30 | M | # | |

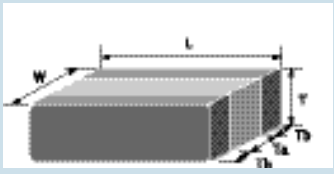
# Reflow soldering only is recommended.
* For safety certificated products, please refer to individual data sheet for detail.

## Capacitor Arrays

| Outline | Size Inch (mm) | L (mm) | W (mm) | T (mm)/Symbol | | S (mm) | BW (mm) | P (mm) |
|---|---|---|---|---|---|---|---|---|
|  | 0612 (1632) | 3.20±0.15 | 1.60±0.15 | 0.80±0.10 | B | 0.30±0.20 | 0.40±0.15 | 0.80±0.15 |

Reflow soldering only.

## Low Inductance Capacitors

| Outline | Size Inch (mm) | L (mm) | W (mm) | T (mm)/Symbol | | $T_a$ min. (mm) | $T_b$ min. (mm) |
|---|---|---|---|---|---|---|---|
|  | 0612 (1632) | 3.20±0.15 | 1.60±0.15 | 0.80±0.10 | B | 0.5 | 0.13 |

Reflow soldering only.

# General Purpose Capacitors

## HOW TO ORDER

| 1206 | F | 104 | Z | 500 | C | T |
|---|---|---|---|---|---|---|
| **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Inch (mm)<br>0402 (1005)<br>0603 (1608)<br>0805 (2012)<br>1206 (3216)<br>1210 (3225)<br>1812 (4532) | N=NP0<br>(C0G)<br>B=X7R<br>F=Y5V | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>R47=0.47pF<br>0R5=0.5pF<br>1R0=1.0pF<br>104=10x10$^4$<br>=100nF | B=±0.1pF<br>C=±0.25pF<br>D=±0.5pF<br>F=±1%<br>G=±2%<br>J=±5%<br>K=±10%<br>M=±20%<br>Z=-20/+80%<br><br>(B,C,D for Cap<10pF; F,G,J,K,M,Z for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>100=10 VDC<br>160=16 VDC<br>250=25 VDC<br>500=50 VDC<br>101=100 VDC | L=Ag/Ni/Sn<br>(for NP0 dielectric)<br><br>C=Cu/Ni/Sn<br>(for X7R, Y5V dielectric) | B=Bulk<br>C=Bulk cassette<br>T=7" reeled<br>G=13" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | | Plastic tape | |
|---|---|---|---|---|---|---|
| | | | 7" reel | 13" reel | 7" reel | 13" reel |
| 0402 (1005) | 0.50±0.05 | N | 10k | 50k | - | - |
| 0603 (1608) | 0.80±0.07 | S | 4k | 15k | - | - |
| | 0.80+0.15/-0.10 | X | 4k | 15k | - | - |
| 0805 (2012) | 0.60±0.10 | A | 4k | 15k | - | - |
| | 0.80±0.10 | B | 4k | 15k | - | - |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| 1206 (3216) | 0.80±0.10 | B | 4k | 15k | - | - |
| | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.15±0.15 | J | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 1.60+0.30/-0.10 | P | - | - | 2k | - |
| 1210 (3225) | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 2.50±0.30 | M | - | - | 1k | - |
| 1812 (4532) | 1.25±0.10 | D | - | - | 1k | - |
| | 2.00±0.20 | K | - | - | 1k | - |

Unit: pieces

# General Purpose Capacitors

## ■ CAPACITANCE RANGE

### NP0 Dielectric

| Dielectric | | NP0 | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | | | 0603 | | | 0805 | | | 1206 | | | 1210 | | | 1812 | |
| Rated Voltage (VDC) | | 16 | 50 | 100 | 16 | 50 | 100 | 16 | 50 | 100 | 16 | 50 | 100 | 16 | 50 | 100 | 50 | 100 |
| Capacitance | 0.5pF (0R5) | | N | N | | S | S | | A | A | | | | | | | | |
| | 0.6pF (0R6) | | N | N | | S | S | | A | A | | | | | | | | |
| | 0.7pF (0R7) | | N | N | | S | S | | A | A | | | | | | | | |
| | 0.8pF (0R8) | | N | N | | S | S | | A | A | | | | | | | | |
| | 0.9pF (0R9) | | N | N | | S | S | | A | A | | | | | | | | |
| | 1.0pF (1R0) | | N | N | | S | S | | A | A | | | | | | | | |
| | 1.2pF (1R2) | | N | N | | S | S | | A | A | | | | | | | | |
| | 1.5pF (1R5) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 1.8pF (1R8) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 2.2pF (2R2) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 2.7pF (2R7) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 3.3pF (3R3) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 3.9pF (3R9) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 4.7pF (4R7) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 5.6pF (5R6) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 6.8pF (6R8) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 8.2pF (8R2) | | N | N | | S | S | | A | A | | B | B | | | | | |
| | 10pF (100) | | N | N | | S | S | | A | A | | B | B | | | C | | D |
| | 12pF (120) | | N | N | | S | S | | A | A | | B | B | | | C | | D |
| | 15pF (150) | | N | N | | S | S | | A | A | | B | B | | | C | | D |
| | 18pF (180) | | N | N | | S | S | | A | A | | B | B | | | C | | D |
| | 22pF (220) | | N | N | | S | S | | A | A | | B | B | | C | C | | D |
| | 27pF (270) | | N | N | | S | S | | A | A | | B | B | | C | C | | D |
| | 33pF (330) | | N | N | | S | S | | A | A | | B | B | | C | C | | D |
| | 39pF (390) | | N | N | | S | S | | A | A | | B | B | | C | C | | D |
| | 47pF (470) | | N | N | | S | S | | A | A | | B | B | | C | C | | D |
| | 56pF (560) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 68pF (680) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 82pF (820) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 100pF (101) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 120pF (121) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 150pF (151) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 180pF (181) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 220pF (221) | | N | | | S | S | | A | A | | B | B | | C | C | | D |
| | 270pF (271) | N | | | | S | S | | A | A | | B | B | | C | C | | D |
| | 330pF (331) | N | | | | S | S | | A | A | | B | B | | C | C | | D |
| | 390pF (391) | N | | | | S | S | | B | B | | B | B | | C | C | | D |
| | 470pF (471) | N | | | | S | S | | B | B | | B | B | | C | C | | D |
| | 560pF (561) | | | | | S | S | | B | B | | B | B | | C | C | | D |
| | 680pF (681) | | | | | S | | | B | B | | B | B | | C | C | | D |
| | 820pF (821) | | | | | S | | | B | B | | B | B | | C | C | | D |
| | 1,000pF (102) | | | | | S | | | B | B | | B | B | | C | C | D | D |
| | 1,200pF (122) | | | | S | | | | B | B | | B | B | | C | C | D | D |
| | 1,500pF (152) | | | | S | | | | B | B | | B | B | | C | C | D | D |
| | 1,800pF (182) | | | | S | | | | B | B | | B | B | | C | C | D | D |
| | 2,200pF (222) | | | | S | | | | B | B | | B | B | | C | C | D | D |
| | 2,700pF (272) | | | | S | | | | D | D | | B | B | | C | C | D | D |
| | 3,300pF (332) | | | | S | | | | D | D | | B | B | | C | C | D | D |
| | 3,900pF (392) | | | | | | | | D | D | | B | B | | C | C | D | D |
| | 4,700pF (472) | | | | | | | | D | | | B | B | | C | C | D | D |
| | 5,600pF (562) | | | | | | | D | | | | B | B | | C | C | D | D |
| | 6,800pF (682) | | | | | | | D | | | | D | D | | C | C | D | D |
| | 8,200pF (822) | | | | | | | D | | | | D | D | | C | C | D | D |
| | 0.010μF (103) | | | | | | | D | | | | D | | | C | C | D | D |
| | 0.012μF (123) | | | | | | | D | | | D | P | | C | D | D | D | D |
| | 0.015μF (153) | | | | | | | | | | D | P | | C | D | D | D | D |
| | 0.018μF (183) | | | | | | | | | | D | | | | | | D | D |
| | 0.022μF (223) | | | | | | | | | | D | | | | | | D | D |
| | 0.027μF (273) | | | | | | | | | | D | | | | | | D | D |
| | 0.033μF (333) | | | | | | | | | | D | | | | | | D | D |
| | 0.039μF (393) | | | | | | | | | | G | | | | | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# General Purpose Capacitors

## X7R Dielectric

| Dielectric | | X7R | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | | | | 0603 | | | | | 0805 | | | | | 1206 | | | | 1210 | | | 1812 | | |
| Rated Voltage (VDC) | | 10 | 16 | 25 | 50 | 10 | 16 | 25 | 50 | 100 | 10 | 16 | 25 | 50 | 100 | 16 | 25 | 50 | 100 | 25 | 50 | 100 | 25 | 50 | 100 |
| Capacitance | 100pF (101) | | | | N | | | | S | S | | | | B | B | | | | | | | | | | |
| | 120pF (121) | | | | N | | | | S | S | | | | B | B | | | | | | | | | | |
| | 150pF (151) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 180pF (181) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 220pF (221) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 270pF (271) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 330pF (331) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 390pF (391) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 470pF (471) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 560pF (561) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 680pF (681) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 820pF (821) | | | | N | | | | S | S | | | | B | B | | | B | B | | | | | | |
| | 1,000pF (102) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 1,200pF (122) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 1,500pF (152) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 1,800pF (182) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 2,200pF (222) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 2,700pF (272) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 3,300pF (332) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 3,900pF (392) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 4,700pF (472) | | | | N | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 5,600pF (562) | | | N | | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 6,800pF (682) | | | N | | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 8,200pF (822) | | | N | | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.010μF (103) | | | N | | | | | S | S | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.012μF (123) | | N | N | | | | | S | | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.015μF (153) | | N | N | | | | | S | | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.018μF (183) | | N | N | | | | | S | | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.022μF (223) | | N | N | | | | | S | | | | | B | B | | | B | B | | C | C | | D | D |
| | 0.027μF (273) | N | | | | | | | S | | | | | B | D | | | B | B | | C | C | | D | D |
| | 0.033μF (333) | N | | | | | | S | X | | | | | B | D | | | B | B | | C | C | | D | D |
| | 0.039μF (393) | N | | | | | | S | X | | | | | B | D | | | B | B | | C | C | | D | D |
| | 0.047μF (473) | N | | | | | | S | X | | | | | B | D | | | B | B | | C | C | | D | D |
| | 0.056μF (563) | N | | | | | | S | X | | | | | B | | | | B | B | | C | C | | D | D |
| | 0.068μF (683) | N | | | | | | S | X | | | | | B | | | | B | B | | C | C | | D | D |
| | 0.082μF (823) | N | | | | | S | S | X | | | | B | B | | | | B | D | | C | C | | D | D |
| | 0.10μF (104) | N | | | | | S | S | X | | | | B | B | | | | B | D | | C | C | | D | D |
| | 0.12μF (124) | | | | | S | S | | | | | | B | D | | | | B | D | | C | C | | D | D |
| | 0.15μF (154) | | | | | S | S | | | | | | D | D | | | | C | G | | C | D | | D | D |
| | 0.18μF (184) | | | | | S | S | | | | | | D | | | | | C | G | | C | D | | D | D |
| | 0.22μF (224) | | | | | S | S | | | | | | D | | | | | C | G | | C | D | | D | D |
| | 0.27μF (274) | | | | | X | | | | | | | D | | | | C | D | | | C | G | | D | D |
| | 0.33μF (334) | | | | | X | | | | | | | D | | | | C | D | | C | D | G | | D | D |
| | 0.39μF (394) | | | | | X | | | | | | D | D | | | C | J | P | | C | D | M | | D | D |
| | 0.47μF (474) | | | | | X | | | | | | D | D | | | J | J | P | | C | D | M | | D | K |
| | 0.56μF (564) | | | | | | | | | | | D | D | | | J | J | P | | D | D | M | | D | K |
| | 0.68μF (684) | | | | | | | | | | D | D | D | | | J | J | P | | D | D | | D | K | K |
| | 0.82μF (824) | | | | | | | | | | D | D | D | | | J | J | P | | D | D | | D | K | K |
| | 1.0μF (105) | | | | | | | | | | D | D | D | | | J | J | P | | D | D | | D | K | K |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# General Purpose Capacitors

## ■ CAPACITANCE RANGE

### Y5V Dielectric (0402,0603,0805,Sizes)

| Dielectric | | Y5V | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | | | | 0603 | | | | 0805 | | | |
| Rated Voltage (VDC) | | 10 | 16 | 25 | 50 | 10 | 16 | 25 | 50 | 16 | 25 | 50 | 100 |
| Capacitance | 0.010μF (103) | | | | N | | | | S | | | A | B |
| | 0.015μF (153) | | | | N | | | | S | | | A | B |
| | 0.022μF (223) | | | | N | | | | S | | | A | B |
| | 0.033μF (333) | | | | N | | | | S | | | A | B |
| | 0.047μF (473) | | | N | | | | | S | | | A | B |
| | 0.068μF (683) | | N | N | | | | | S | | | A | B |
| | 0.10μF (104) | | N | N | | | | | S | | | A | B |
| | 0.15μF (154) | N | | | | | | | S | | | A | B |
| | 0.22μF (224) | N | | | | | | S | | | | A | |
| | 0.33μF (334) | | | | | | | S | | | | B | |
| | 0.47μF (474) | | | | | | S | | | | B | B | |
| | 0.68μF (684) | | | | | S | X | | | B | D | | |
| | 1.0μF (105) | | | | | S | X | | | B | D | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

### Y5V Dielectric (1206,1210,1812 Sizes)

| Dielectric | | Y5V | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Size | | 1206 | | | 1210 | | 1812 | |
| Rated Voltage (VDC) | | 25 | 50 | 100 | 50 | 100 | 50 | 100 |
| Capacitance | 0.010μF (103) | B | B | B | | C | | D |
| | 0.015μF (153) | B | B | B | | C | | D |
| | 0.022μF (223) | B | B | B | | C | | D |
| | 0.033μF (333) | B | B | B | | C | | D |
| | 0.047μF (473) | B | B | B | | C | | D |
| | 0.068μF (683) | B | B | B | | C | | D |
| | 0.10μF (104) | B | B | B | C | C | D | D |
| | 0.15μF (154) | B | B | C | C | C | D | D |
| | 0.22μF (224) | B | B | C | C | C | D | D |
| | 0.33μF (334) | B | B | | C | C | D | D |
| | 0.47μF (474) | B | B | | C | | D | D |
| | 0.68μF (684) | B | B | | C | | D | D |
| | 1.0μF (105) | C | C | | C | | D | D |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# High Capacitance Capacitors

## HOW TO ORDER

| 1206 | F | 106 | Z | 100 | C | T |
|---|---|---|---|---|---|---|
| **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Inch (mm)<br>0402 (1005)<br>0603 (1608)<br>0805 (2012)<br>1206 (3216)<br>1210 (3225)<br>1812 (4532) | B=X7R<br>X=X5R<br>S=X6S<br>F=Y5V | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>$106=10x10^6$<br>=10μF | K=±10%<br>M=±20%<br>Z=-20/+80% | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>6R3=6.3 VDC<br>100=10 VDC<br>160=16 VDC<br>250=25 VDC<br>500=50 VDC | C=Cu/Ni/Sn | B=Bulk<br>C=Bulk cassette<br>T=7" reeled<br>G=13" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | | Plastic tape | |
|---|---|---|---|---|---|---|
| | | | 7" reel | 13" reel | 7" reel | 13" reel |
| 0402 (1005) | 0.50±0.05 | N | 10k | 50k | - | - |
| 0603 (1608) | 0.80±0.07 | S | 4k | 15k | - | - |
| | 0.80+0.15/-0.10 | X | 4k | 15k | - | - |
| 0805 (2012) | 0.80±0.10 | B | 4k | 15k | - | - |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.25±0.20 | I | - | - | 3k | 10k |
| 1206 (3216) | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.15±0.15 | J | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 1.60+0.30/-0.10 | P | - | - | 2k | - |
| 1210 (3225) | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 2.00±0.20 | K | - | - | 1k | - |
| | 2.50±0.30 | M | - | - | 1k | - |
| 1812 (4532) | 1.25±0.10 | D | - | - | 1k | - |
| | 2.00±0.20 | K | - | - | 1k | - |
| | 2.50±0.30 | M | - | - | 0.5k | - |

Unit: pieces

# High Capacitance Capacitors

## ■ CAPACITANCE RANGE

### X7R Dielectric

| Dielectric | | X7R | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | 0603 | | | | 0805 | | | 1206 | | | | 1210 | | 1812 | |
| Rated Voltage (VDC) | | 10 | 10 | 16 | 25 | 50 | 10 | 16 | 25 | 10 | 16 | 25 | 50 | 25 | 50 | 25 | 50 |
| Capacitance | 0.10μF (104) | N | | S | S | X | | | | | | | | | | | |
| | 0.15μF (154) | | S | S | | | | | | | | | | | | | |
| | 0.22μF (224) | | S | S | | | | | | | | | | | | | |
| | 0.33μF (334) | | X | X | | | | | | | | | | | | | |
| | 0.47μF (474) | | X | X | | | | | | | | | | | | | |
| | 0.68μF (684) | | | | | | | | | | | | | | | | |
| | 1.0μF (105) | | X | | | | D | D | D | | J | J | P | D | D | D | K |
| | 1.5μF (155) | | | | | | | | | | | | | | | | |
| | 2.2μF (225) | | | | | | | | | J | J | P | | G | | | |
| | 3.3μF (335) | | | | | | | | | P | | | | | | | |
| | 4.7μF (475) | | | | | | | | | P | | | | | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

### X5R Dielectric

| Dielectric | | X5R | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | | | 0603 | | | 0805 | | 1206 | | | 1210 | |
| Rated Voltage (VDC) | | 6.3 | 10 | 16 | 6.3 | 10 | 16 | 6.3 | 10 | 6.3 | 10 | 16 | 10 | 16 |
| Capacitance | 0.027μF (273) | | | N | | | | | | | | | | |
| | 0.033μF (333) | | | N | | | | | | | | | | |
| | 0.039μF (393) | | | N | | | | | | | | | | |
| | 0.047μF (473) | | | N | | | | | | | | | | |
| | 0.056μF (563) | | N | N | | | | | | | | | | |
| | 0.068μF (683) | | N | N | | | | | | | | | | |
| | 0.082μF (823) | | N | N | | | | | | | | | | |
| | 0.10μF (104) | | N | N | | | | | | | | | | |
| | 0.15μF (154) | N | | | | | | | | | | | | |
| | 0.22μF (224) | N | | | | | | | | | | | | |
| | 0.33μF (334) | | | | | X | X | | | | | | | |
| | 0.47μF (474) | | | | | X | X | | | | | | | |
| | 0.68μF (684) | | | | X | X | X | | | | | | | |
| | 1.0μF (105) | | | | X | X | | | | | | | | |
| | 1.5μF (155) | | | | | | | | | | | J | | |
| | 2.2μF (225) | | | | | | | I | I | | J | J | | |
| | 3.3μF (335) | | | | | | | | | | P | P | | |
| | 4.7μF (475) | | | | | | | I | | | P | P | | |
| | 6.8μF (685) | | | | | | | | | P | P | | | |
| | 10μF (106) | | | | | | | I | | P | P | | K | K |

1. The letter in cell is expressed the symbol of product thickness.

### Y5V Dielectric

| Dielectric | | Y5V | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0402 | | 0603 | | 0805 | | | 1206 | | | | | 0805 | | 1210 | | | | 1812 | | |
| Rated Voltage (VDC) | | 6.3 | 10 | 10 | 16 | 10 | 16 | 25 | 10 | 16 | 25 | 35 | 50 | 6.3 | 10 | 16 | 25 | 35 | 50 | 16 | 25 | 50 |
| Capacitance | 0.15μF (154) | | N | | | | | | | | | | | | | | | | | | | |
| | 0.22μF (224) | | N | | | | | | | | | | | | | | | | | | | |
| | 0.33μF (334) | N | | | | | | | | | | | | | | | | | | | | |
| | 0.47μF (474) | N | | | | | | | | | | | | | | | | | | | | |
| | 0.68μF (684) | N | | | | | | | | | | | | | | | | | | | | |
| | 1.0μF (105) | N | | S | X | B | B | D | | C | C | | C | | | | | | C | | | D |
| | 1.5μF (155) | | | S | | D | D | | | C | C | | | | | | C | | | | | D |
| | 2.2μF (225) | | | S | | D | D | | | C | C | | | | | | C | | | | | D |
| | 3.3μF (335) | | | | | D | D | | J | J | J | | | | | | C | | | | | D |
| | 4.7μF (475) | | | | | D | D | | J | J | J | J | | | | C | D | | | | | D |
| | 6.8μF (685) | | | | | I | | | J | J | | | | | | C | G | | | | | D |
| | 10μF (106) | | | | | I | | | J | J | | | | | | D | G | K | | | D | |
| | 22μF (226) | | | | | | | | P | | | | | | K | K | | | | | | |
| | 47μF (476) | | | | | | | | | | | | | | K | | | | | | | |
| | 100μF (107) | | | | | | | | | | | | | M | | | | | | M | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Low Profile Capacitors

## HOW TO ORDER

| TT | 31 | X | 225 | M | 100 | C | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| TT=Low profile | 21=0805 (2012)<br>31=1206 (3216)<br>32=1210 (3225) | X=X5R<br>F=Y5V | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>225=22x10$^5$<br>=2,200,000pF<br>=2.2μF | K=±10%<br>M=±20%<br>Z=+20/-80% | Two significant digits followed by no. of zeros.And R is in place of decimal point.<br><br>6R3=6.3 VDC<br>100=10 VDC<br>160=16 VDC | C=Cu/Ni/Sn | B=Bulk<br>T=7" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness Max (mm)/Symbol | | 7" reel | |
|---|---|---|---|---|
| | | | Paper tape | Plastic tape |
| 0805 (2012) | 0.95 | T | 4k | - |
| 1206 (3216) | 0.95 | T | 4k | - |
| | 1.25 | J | - | 3k |
| 1210 (3225) | 0.95 | T | - | 3k |

Unit: pieces

## CAPACITANCE RANGE

| Dielectric | Size | Capacitance | Tolerance | Rated Voltage (VDC) | Thickness Max (mm) | Part Number |
|---|---|---|---|---|---|---|
| X5R | 0805 | 1.0μF | ±10%, ±20% | 10 | 0.95 | TT21X105□100CT |
| | | 4.7μF | ±10%, ±20% | 6.3 | 0.95 | TT21X475□6R3CT |
| | 1206 | 2.2μF | ±10%, ±20% | 10 | 0.95 | TT31X225□100CT |
| | 1210 | 3.3μF | ±10%, ±20% | 10 | 0.95 | TT32X335□100CT |
| | | 4.7μF | ±10%, ±20% | 10 | 0.95 | TT32X475□100CT |
| Y5V | 0805 | 2.2μF | -20/+80% | 16 | 0.95 | TT21F225Z160CT |
| | | 3.3μF | -20/+80% | 10 | 0.95 | TT21F335Z100CT |
| | | 4.7μF | -20/+80% | 10 | 0.95 | TT21F475Z100CT |
| | 1206 | 4.7μF | -20/+80% | 16 | 0.95 | TT31F475Z160CT |
| | | 10μF | -20/+80% | 10 | 0.95 | TT31F106Z100CT |
| | | 10μF | -20/+80% | 16 | 1.25 | TT31F106Z160CT |
| | 1210 | 10μF | -20/+80% | 10 | 0.95 | TT32F106Z100CT |

□ Please specify the capacitance tolerance code.
1. This series product is suited to reflow soldering process only.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Ultra-small 0201 Capacitors

## HOW TO ORDER

| 0201 | N | 100 | J | 250 | L | T |
|---|---|---|---|---|---|---|
| **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Inch (mm)<br>0201 (0603) | N=NP0<br>(C0G)<br>B=X7R<br>X=X5R | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>R47=0.47pF<br>0R5=0.5pF<br>1R0=1.0pF<br>100=10x10⁰<br>=10pF | B=±0.1pF<br>C=±0.25pF<br>D=±0.5pF<br>G=±2%<br>J=±5%<br>K=±10%<br>M=±20%<br><br>(B,C,D for Cap<10pF; G,J,K,M for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>6R3=6.3 VDC<br>100=10 VDC<br>160=16 VDC<br>250=25 VDC<br>500=50 VDC | L=Ag/Ni/Sn<br>(for NP0 dielectric)<br>C=Cu/Ni/Sn (for X7R, X5R, dielectric) | T=7" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | |
|---|---|---|---|---|
| | | | 7" reel | 13" reel |
| 0201 (0603) | 0.30±0.03 | L | 15K | - |

Unit: pieces

## CAPACITANCE RANGE

| Size | | 0201 | |
|---|---|---|---|
| Dielectric | | NP0 | |
| Rated Voltage (VDC) | | 16 | 25 |
| Capacitance | 0.3pF (0R3) | | L |
| | 0.4pF (0R4) | | L |
| | 0.5pF (0R5) | | L |
| | 1.0pF (1R0) | | L |
| | 1.2pF (1R2) | | L |
| | 1.5pF (1R5) | | L |
| | 1.8pF (1R8) | | L |
| | 2.2pF (2R2) | | L |
| | 2.7pF (2R7) | | L |
| | 3.3pF (3R3) | | L |
| | 3.9pF (3R9) | | L |
| | 4.7pF (4R7) | | L |
| | 5.6pF (5R6) | | L |
| | 6.8pF (6R8) | | L |
| | 8.2pF (8R2) | | L |
| | 10pF (100) | | L |
| | 12pF (120) | | L |
| | 15pF (150) | | L |
| | 18pF (180) | | L |
| | 22pF (220) | | L |
| | 27pF (270) | | L |
| | 33pF (330) | | L |
| | 39pF (390) | | L |
| | 47pF (470) | | L |
| | 56pF (560) | L | L |
| | 68pF (680) | L | L |
| | 82pF (820) | L | L |
| | 100pF (101) | L | L |

| Size | | 0201 | | | | | |
|---|---|---|---|---|---|---|---|
| Dielectric | | X7R | | | X5R | | |
| Rated Voltage (VDC) | | 16V | 25V | 50V | 6.3V | 10V | 16V |
| Capacitance | 100pF (101) | L | L | L | | | |
| | 120pF (121) | L | L | L | | | |
| | 150pF (151) | L | L | L | | | |
| | 180pF (181) | L | L | L | | | |
| | 220pF (221) | L | L | L | | | |
| | 270pF (271) | L | L | L | | | |
| | 330pF (331) | L | L | L | | | |
| | 390pF (391) | L | L | L | | | |
| | 470pF (471) | L | L | L | | | |
| | 560pF (561) | L | L | L | | | |
| | 680pF (681) | L | L | L | | | |
| | 820pF (821) | L | L | L | | | |
| | 1,000pF (102) | L | L | L | | | |
| | 1,500pF (152) | L | | | | | L |
| | 2,200pF (222) | L | | | | | L |
| | 3,300pF (332) | L | | | | | L |
| | 4,700pF (472) | L | | | | | L |
| | 6,800pF (682) | | | | | L | |
| | 0.010μF (103) | | | | | L | |
| | 0.015μF (153) | | | | L | | |
| | 0.022μF (223) | | | | L | | |
| | 0.033μF (333) | | | | | | |
| | 0.047μF (473) | | | | | | |
| | 0.068μF (683) | | | | | | |
| | 0.10μF (104) | | | | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local sales representative.

# Middle and High Voltage Capacitors

## HOW TO ORDER

| 1808 | N | 100 | G | 202 | L | T |
|---|---|---|---|---|---|---|
| **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Inch (mm)<br>0603(1608)<br>0805 (2012)<br>1206 (3216)<br>1210 (3225)<br>1808 (4520)<br>1812 (4532) | N=NP0<br>(C0G)<br>B=X7R<br>F=Y5V | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>R47=0.47pF<br>0R5=0.5pF<br>1R0=1.0pF<br>100=10x10⁰<br>=10pF | B=±0.1pF<br>C=±0.25pF<br>D=±0.5pF<br>G=±2%<br>J=±5%<br>K=±10%<br>M=±20%<br>Z=-20/+80%<br><br>(B,C,D for Cap<10pF; G,J,K,M,Z for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>201=200 VDC<br>251=250 VDC<br>501=500 VDC<br>631=630 VDC<br>102=1000 VDC<br>152=1500 VDC<br>202=2000 VDC<br>302=3000 VDC | L=Ag/Ni/Sn<br>(for NP0 dielectric)<br><br>C=Cu/Ni/Sn<br>(for X7R*, Y5V dielectric) | B=Bulk<br>T=7" reeled<br>G=13" reeled |

* Partial X7R items are with Ag/Ni/Sn terminations, please ref to below product range of X7R dielectric for detail.

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | | Plastic tape | |
|---|---|---|---|---|---|---|
| | | | 7" reel | 13" reel | 7" reel | 13" reel |
| 0603 (1608) | 0.80±0.07 | S | 4k | 15k | - | - |
| 0805 (2012) | 0.60±0.10 | A | 4k | 15k | - | - |
| | 0.80±0.10 | B | 4k | 15k | - | - |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| 1206 (3216) | 0.80±0.10 | B | 4k | 15k | - | - |
| | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| 1210 (3225) | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 2.50±0.30 | M | - | - | 1k | - |
| 1808 (4520) | 1.25±0.10 | D | - | - | 2k | - |
| | 2.00±0.20 | K | - | - | 1k | - |
| 1812 (4532) | 1.25±0.10 | D | - | - | 1k | - |
| | 2.00±0.20 | K | - | - | 1k | - |

Unit: pieces

# Middle and High Voltage Capacitors

## ■ CAPACITANCE RANGE

### NP0 Dielectric 200V to 630V

| Dielectric | | NP0 | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0603 | | 0805 | | | | 1206 | | | | 1210 | | | | 1812 | | | |
| Rated Voltage (VDC) | | 200 | 250 | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 |
| Capacitance | 0.5pF (0R5) | | | A | A | A | A | | | | | | | | | | | | |
| | 1.0pF (1R0) | | | A | A | A | A | | | | | | | | | | | | |
| | 1.2pF (1R2) | | | A | A | A | A | | | | | | | | | | | | |
| | 1.5pF (1R5) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 1.8pF (1R8) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 2.2pF (2R2) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 2.7pF (2R7) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 3.3pF (3R3) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 3.9pF (3R9) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 4.7pF (4R7) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 5.6pF (5R6) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 6.8pF (6R8) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 8.2pF (8R2) | | | A | A | A | A | B | B | B | B | | | | | | | | |
| | 10pF (100) | | | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 12pF (120) | | | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 15pF (150) | | | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 18pF (180) | | | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 22pF (220) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 27pF (270) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 33pF (330) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 39pF (390) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 47pF (470) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 56pF (560) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 68pF (680) | S | S | A | A | A | A | B | B | B | B | C | C | C | C | D | D | D | D |
| | 82pF (820) | S | S | A | A | B | B | B | B | B | B | C | C | C | C | D | D | D | D |
| | 100pF (101) | S | S | A | B | B | B | B | B | B | B | C | C | C | C | D | D | D | D |
| | 120pF (121) | | | A | B | D | D | B | B | B | B | C | C | C | C | D | D | D | D |
| | 150pF (151) | | | B | D | D | D | B | B | B | B | C | C | C | C | D | D | D | D |
| | 180pF (181) | | | B | D | D | D | B | B | B | B | C | C | C | C | D | D | D | D |
| | 220pF (221) | | | D | D | D | D | B | B | B | B | C | C | C | C | D | D | D | D |
| | 270pF (271) | | | D | D | D | D | B | C | C | C | C | C | C | C | D | D | D | D |
| | 330pF (331) | | | D | D | D | D | B | C | C | C | C | C | C | C | D | D | D | D |
| | 390pF (391) | | | D | D | D | D | B | C | C | C | C | C | C | C | D | D | D | D |
| | 470pF (471) | | | D | | | | C | C | C | C | C | C | C | C | D | D | D | D |
| | 560pF (561) | | | D | | | | C | D | D | D | C | C | C | C | D | D | D | D |
| | 680pF (681) | | | D | | | | C | D | D | D | C | C | C | C | D | D | D | D |
| | 820pF (821) | | | D | | | | C | G | G | G | C | C | C | C | D | D | D | D |
| | 1,000pF (102) | | | | | | | C | G | G | G | C | D | D | D | D | D | D | D |
| | 1,200pF (122) | | | | | | | C | | | | C | D | D | D | D | D | D | D |
| | 1,500pF (152) | | | | | | | D | | | | D | D | D | D | D | D | D | D |
| | 1,800pF (182) | | | | | | | D | | | | D | D | D | D | D | D | D | D |
| | 2,200pF (222) | | | | | | | D | | | | D | D | | | D | D | D | D |
| | 2,700pF (272) | | | | | | | | | | | D | D | | | D | D | D | D |
| | 3,300pF (332) | | | | | | | | | | | D | | | | D | D | D | D |
| | 3,900pF (392) | | | | | | | | | | | D | | | | D | | | |
| | 4,700pF (472) | | | | | | | | | | | | | | | D | | | |
| | 5,600pF (562) | | | | | | | | | | | | | | | D | | | |
| | 6,800pF (682) | | | | | | | | | | | | | | | D | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Middle and High Voltage Capacitors

## NP0 Dielectric 1kV to 3kV

| Dielectric | | NP0 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 1206 | | 1210 | | 1808 | | | 1812 | | |
| Rated Voltage (VDC) | | 1000 | 2000 | 1000 | 2000 | 1000 | 2000 | 3000 | 1000 | 2000 | 3000 |
| Capacitance | 1.5pF (1R5) | B | B | | | | | | | | |
| | 1.8pF (1R8) | B | B | | | | | | | | |
| | 2.2pF (2R2) | B | B | | | D | D | D | | | |
| | 2.7pF (2R7) | B | B | | | D | D | D | | | |
| | 3.3pF (3R3) | B | B | | | D | D | D | | | |
| | 3.9pF (3R9) | B | B | | | D | D | D | | | |
| | 4.7pF (4R7) | B | B | | | D | D | D | | | |
| | 5.6pF (5R6) | B | B | | | D | D | D | | | |
| | 6.8pF (6R8) | B | B | | | D | D | D | | | |
| | 8.2pF (8R2) | B | B | | | D | D | D | | | |
| | 10pF (100) | B | B | C | C | D | D | D | D | D | D |
| | 12pF (120) | B | B | C | C | D | D | D | D | D | D |
| | 15pF (150) | B | B | C | C | D | D | D | D | D | D |
| | 18pF (180) | B | B | C | C | D | D | D | D | D | D |
| | 22pF (220) | B | B | C | C | D | D | D | D | D | D |
| | 27pF (270) | B | B | C | C | D | D | D | D | D | D |
| | 33pF (330) | B | B | C | C | D | D | D | D | D | D |
| | 39pF (390) | B | B | C | C | D | D | D | D | D | D |
| | 47pF (470) | B | B | C | C | D | D | D | D | D | D |
| | 56pF (560) | B | B | C | D | D | D | D | D | D | D |
| | 68pF (680) | B | C | C | D | D | D | D | D | D | D |
| | 82pF (820) | B | C | C | D | D | D | D | D | D | D |
| | 100pF (101) | B | C | C | D | D | D | D | D | D | D |
| | 120pF (121) | B | D | C | D | D | D | D | D | D | D |
| | 150pF (151) | C | D | C | D | D | D | D | D | D | D |
| | 180pF (181) | C | G | C | D | D | D | K | D | D | D |
| | 220pF (221) | D | G | C | D | D | D | K | D | D | D |
| | 270pF (271) | D | | C | | D | D | K | D | D | K |
| | 330pF (331) | G | | D | | D | D | | D | D | K |
| | 390pF (391) | G | | D | | D | K | | D | D | K |
| | 470pF (471) | G | | D | | D | K | | D | D | K |
| | 560pF (561) | | | | | K | K | | D | D | |
| | 680pF (681) | | | | | K | K | | D | K | |
| | 820pF (821) | | | | | K | | | D | K | |
| | 1,000pF (102) | | | | | K | | | K | K | |
| | 1,200pF (122) | | | | | | | | K | | |
| | 1,500pF (152) | | | | | | | | K | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Middle and High Voltage Capacitors

## X7R Dielectric 200V to 630V

| Dielectric | | X7R | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 0805 | | | | 1206 | | | | 1210 | | | | 1812 | | | |
| Rated Voltage (VDC) | | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 | 200 | 250 | 500 | 630 |
| Capacitance | 100pF (101) | B | B | B^ | B^ | | | | | | | | | | | | |
| | 120pF (121) | B | B | B^ | B^ | | | | | | | | | | | | |
| | 150pF (151) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 180pF (181) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 220pF (221) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 270pF (271) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 330pF (331) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 390pF (391) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 470pF (471) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 560pF (561) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 680pF (681) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 820pF (821) | B | B | B^ | B^ | B | B | B^ | B^ | | | | | | | | |
| | 1,000pF (102) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 1,200pF (122) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 1,500pF (152) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 1,800pF (182) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 2,200pF (222) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 2,700pF (272) | B | B | B^ | B^ | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 3,300pF (332) | B | B | | | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 3,900pF (392) | B | B | | | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 4,700pF (472) | B | B | | | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 5,600pF (562) | D | D | | | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 6,800pF (682) | D | D | | | B | B | B^ | B^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 8,200pF (822) | D | D | | | B | B | C^ | C^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 0.010μF (103) | D | D | | | B | B | C^ | C^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 0.012μF (123) | D | D | | | B | B | D^ | D^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 0.015μF (153) | D | D | | | C | C | D^ | D^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 0.018μF (183) | D | D | | | C | C | D^ | D^ | C | C | C^ | C^ | D | D | D^ | D^ |
| | 0.022μF (223) | D | D | | | C | C | G^ | G^ | C | C | D^ | D^ | D | D | D^ | D^ |
| | 0.027μF (273) | | | | | C | C | G^ | G^ | C | C | G^ | G^ | D | D | D^ | D^ |
| | 0.033μF (333) | | | | | G | G | G^ | G^ | C | C | G^ | G^ | D | D | D^ | D^ |
| | 0.039μF (393) | | | | | G | G | | | C | C | G^ | G^ | D | D | D^ | D^ |
| | 0.047μF (473) | | | | | G | G | | | D | D | G^ | G^ | D | D | D^ | D^ |
| | 0.056μF (563) | | | | | G | G | | | D | D | G^ | G^ | D | D | K^ | K^ |
| | 0.068μF (683) | | | | | G | G | | | G | G | | | D | D | K^ | K^ |
| | 0.082μF (823) | | | | | G | G | | | G | G | | | D | D | K^ | K^ |
| | 0.10μF (104) | | | | | G | G | | | G | G | | | D | D | K^ | K^ |
| | 0.12μF (124) | | | | | | | | | G | G | | | D | D | | |
| | 0.15μF (154) | | | | | | | | | M | M | | | K | K | | |
| | 0.18μF (184) | | | | | | | | | M | M | | | K | K | | |
| | 0.22μF (224) | | | | | | | | | M | M | | | K | K | | |
| | 0.27μF (274) | | | | | | | | | | | | | K | K | | |
| | 0.33μF (334) | | | | | | | | | | | | | K | K | | |
| | 0.39μF (394) | | | | | | | | | | | | | K | K | | |
| | 0.47μF (474) | | | | | | | | | | | | | K | K | | |

1. The letter in cell is expressed the symbol of product thickness.
2. The letter in cell with "^" mark is expressed product with Ag/Ni/Sn terminations.
3. For more information about products with special capacitance or other data, please contact WTC local representative.

# Middle and High Voltage Capacitors

## X7R Dielectric 1kV to 3kV

| Dielectric | | X7R | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | | 1206 | | | 1210 | 1808 | | | | 1812 | | |
| Rated Voltage (VDC) | | 1000 | 1500 | 2000 | 1000 | 1000 | 1500 | 2000 | 3000 | 1000 | 2000 | 3000 |
| Capacitance | 150pF (151) | B^ | B^ | B^ | | D^ | D^ | D^ | | | | |
| | 180pF (181) | B^ | B^ | B^ | | D^ | D^ | D^ | | | | |
| | 220pF (221) | B^ | B^ | B^ | | D^ | D^ | D^ | | | | |
| | 270pF (271) | B^ | B^ | B^ | | D^ | D^ | D^ | | D^ | D^ | |
| | 330pF (331) | B^ | B^ | B^ | | D^ | D^ | D^ | | D^ | D^ | |
| | 390pF (391) | B^ | B^ | C^ | | D^ | D^ | D^ | | D^ | D^ | |
| | 470pF (471) | B^ | B^ | C^ | | D^ | D^ | D^ | K^ | D^ | D^ | |
| | 560pF (561) | B^ | C^ | C^ | | D^ | D^ | D^ | K^ | D^ | D^ | |
| | 680pF (681) | B^ | C^ | C^ | | D^ | D^ | D^ | K^ | D^ | D^ | K^ |
| | 820pF (821) | B^ | G^ | G^ | | D^ | D^ | D^ | K^ | D^ | D^ | K^ |
| | 1,000pF (102) | B^ | G^ | G^ | C^ | D^ | D^ | K^ | K^ | D^ | D^ | K^ |
| | 1,200pF (122) | B^ | G^ | | C^ | D^ | D^ | K^ | | D^ | D^ | |
| | 1,500pF (152) | C^ | G^ | | C^ | D^ | D^ | K^ | | D^ | D^ | |
| | 1,800pF (182) | C^ | G^ | | C^ | D^ | D^ | K^ | | D^ | D^ | |
| | 2,200pF (222) | D^ | G^ | | C^ | D^ | D^ | K^ | | D^ | D^ | |
| | 2,700pF (272) | G^ | | | C^ | D^ | D^ | | | D^ | D^ | |
| | 3,300pF (332) | G^ | | | D^ | D^ | K^ | | | D^ | K^ | |
| | 3,900pF (392) | G^ | | | | D^ | | | | D^ | K^ | |
| | 4,700pF (472) | | | | | D^ | | | | D^ | K^ | |
| | 5,600pF (562) | | | | | K^ | | | | D^ | | |
| | 6,800pF (682) | | | | | K^ | | | | D^ | | |
| | 8,200pF (822) | | | | | K^ | | | | D^ | | |
| | 0.010μF (103) | | | | | K^ | | | | D^ | | |
| | 0.012μF (123) | | | | | | | | | K^ | | |
| | 0.015μF (153) | | | | | | | | | K^ | | |

1. The letter in cell is expressed the symbol of product thickness.
2. The letter in cell with "^" mark is expressed product with Ag/Ni/Sn terminations.
3. For more information about products with special capacitance or other data, please contact WTC local representative.

## Y5V Dielectric 200V to 250V

| Dielectric | | Y5V | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Size | | 0805 | | 1206 | | 1210 | | 1812 | |
| Rated Voltage (VDC) | | 200 | 250 | 200 | 250 | 200 | 250 | 200 | 250 |
| Capacitance | 0.010μF (103) | B | B | B | B | C | C | D | D |
| | 0.015μF (153) | B | B | B | B | C | C | D | D |
| | 0.022μF (223) | B | B | B | B | C | C | D | D |
| | 0.033μF (333) | B | B | B | B | C | C | D | D |
| | 0.047μF (473) | B | B | B | B | C | C | D | D |
| | 0.068μF (683) | B | B | B | B | C | C | D | D |
| | 0.10μF (104) | | | B | B | C | C | D | D |
| | 0.15μF (154) | | | C | C | C | C | D | D |
| | 0.22μF (224) | | | | | | | D | D |
| | 0.33μF (334) | | | | | | | D | D |
| | 0.47μF (474) | | | | | | | D | D |
| | 0.68μF (684) | | | | | | | D | D |
| | 1.0μF (105) | | | | | | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Safety Certificated Capacitors X1/Y2

## HOW TO ORDER

| S2 | 42 | N | 100 | J | 302 | L | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| S2=X1/Y2 | 42=1808 (4520)<br><br>43=1812 (4532) | N=NP0 (C0G) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>100=10x10⁰ =10pF | J=±5%<br>K=±10% | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>302=3000 VDC | L=Ag/Ni/Sn | B=Bulk<br>T=7" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | 7" reel / Plastic tape |
|---|---|---|---|
| 1808 (4520) | 1.25±0.10 | D | 2k |
| | 2.00±0.20 | K | 1k |
| 1812 (4532) | 1.25±0.10 | D | 1k |
| | 2.00±0.20 | K | 1k |

Unit: pieces

## CAPACITANCE RANGE

| Dielectric | | NP0 | |
|---|---|---|---|
| Rated voltage (VAC) | | 250 | |
| Rated Voltage (VDC) | | 3000 | |
| Size | | 1808 | 1812 |
| Capacitance | 10pF (100) | D | |
| | 12pF (120) | D | D |
| | 15pF (150) | D | D |
| | 18pF (180) | D | D |
| | 22pF (220) | D | D |
| | 27pF (270) | D | D |
| | 33pF (330) | D | D |
| | 39pF (390) | D | D |
| | 47pF (470) | D | D |
| | 56pF (560) | D | D |
| | 68pF (680) | D | D |
| | 82pF (820) | D | D |
| | 100pF (101) | D | D |
| | 120pF (121) | D | D |
| | 150pF (151) | D | D |
| | 180pF (181) | K | D |
| | 220pF (221) | K | D |
| | 270pF (271) | K | K |
| | 330pF (331) | | K |
| | 390pF (391) | | K |
| | 470pF (471) | | K |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Safety Certificated Capacitors X2/Y3

## HOW TO ORDER

| S3 | 42 | N | 100 | J | 202 | L | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| S3=X2/Y3 | 42=1808 (4520)<br><br>43=1812 (4532) | N=NP0 (C0G)<br>B=X7R | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>Eg.:<br>R47=4.7pF<br>0R5=0.5pF<br>1R0=1.0pF<br>100=10x10⁰ =10pF | C=±0.25pF<br>D=±0.5pF<br>J=±5%<br>K=±10%<br><br>(C,D for Cap<10pF; J,K for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>202=2000 VDC<br>302=3000 VDC | L=Ag/Ni/Sn | B=Bulk<br>T=7" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | 7" reel / Plastic tape |
|---|---|---|---|
| 1808 (4520) | 1.25±0.10 | D | 2k |
| | 2.00±0.20 | K | 1k |
| 1812 (4532) | 1.25±0.10 | D | 1k |
| | 2.00±0.20 | K | 1k |

Unit: pieces

# Safety Certificated Capacitors X2/Y3

## ■ CAPACITANCE RANGE

### NP0 Dielectric

| Dielectric | | NP0 | | | |
|---|---|---|---|---|---|
| Size | | 1808 | | 1812 | |
| Rated voltage (VAC) | | 250 | | 250 | |
| Rated Voltage (VDC) | | 2000 | 3000 | 2000 | 3000 |
| Capacitance | 3.9pF (3R9) | | D* | | |
| | 4.7pF (4R7) | | D* | | |
| | 5.0pF (5R0) | | D* | | |
| | 5.6pF (5R6) | | D* | | |
| | 6.8pF (6R8) | | D* | | |
| | 8.2pF (8R2) | | D* | | |
| | 10pF (100) | D | D | D* | D* |
| | 12pF (120) | D | D | D | D |
| | 15pF (150) | D | D | D | D |
| | 18pF (180) | D | D | D | D |
| | 22pF (220) | D | D | D | D |
| | 27pF (270) | D | D | D | D |
| | 33pF (330) | D | D | D | D |
| | 39pF (390) | D | D | D | D |
| | 47pF (470) | D | D | D | D |
| | 56pF (560) | D | D | D | D |
| | 68pF (680) | D | D | D | D |
| | 82pF (820) | D | D | D | D |
| | 100pF (101) | D | D | D | D |
| | 120pF (121) | D | D | D | D |
| | 150pF (151) | D | D | D | D |
| | 180pF (181) | D | K | D | D |
| | 220pF (221) | D | K | D | D |
| | 270pF (271) | D | K | D | K |
| | 330pF (331) | D | | D | K |
| | 390pF (391) | K | | D | K |
| | 470pF (471) | K | | D | K |
| | 560pF (561) | K | | D | |
| | 680pF (681) | K | | K | |
| | 820pF (821) | | | K | |
| | 1,000pF (102) | | | K | |

"*"means it is only available for UL safety certificated.
1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

### X7R Dielectric

| Dielectric | | X7R | | | |
|---|---|---|---|---|---|
| Size | | 1808 | | 1812 | |
| Rated voltage (VAC) | | 250 | | 250 | |
| Rated Voltage (VDC) | | 2000 | 3000 | 2000 | 3000 |
| Capacitance | 150pF (151) | D | | | |
| | 180pF (181) | D | | | |
| | 220pF (221) | D | | | |
| | 270pF (271) | D | | D | |
| | 330pF (331) | D | K* | D | |
| | 390pF (391) | D | K* | D | |
| | 470pF (471) | D | K* | D | |
| | 560pF (561) | D | K | D | |
| | 680pF (681) | D | K | D | K |
| | 820pF (821) | D | K | D | K |
| | 1,000pF (102) | K | K | D | K |
| | 1,200pF (122) | K | | D | |
| | 1,500pF (152) | K | | D | |
| | 1,800pF (182) | K | | D | |
| | 2,200pF (222) | K | | D | |
| | 2,700pF (272) | | | D | |
| | 3,300pF (332) | | | K | |
| | 3,900pF (392) | | | K | |
| | 4,700pF (472) | | | K | |

"*"means it is only available for UL safety certificated.
1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# High Q and Low ESR Capacitors

## HOW TO ORDER

| HH | 15 | N | 100 | G | 500 | L | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| HH=<br>High Q/ Low ESR | 15=0402<br>(1005)<br><br>18=0603<br>(1608) | N=NP0<br>(C0G) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>R47=0.47pF<br>0R5=0.5pF<br>1R0=1.0pF<br>100=10x10⁰<br>=10pF | B=±0.1pF<br>C=±0.25pF<br>D=±0.5pF<br>F=±1%<br>G=±2%<br>J=±5%<br><br>(B,C,D for Cap<10pF ; F,G,J for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>160=16 VDC<br>250=25 VDC<br>500=50 VDC<br>101=100 VDC | L=Ag/Ni/Sn | B=Bulk<br>C=Bulk cassette<br>T=7" reeled<br>G=13" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | |
|---|---|---|---|---|
| | | | 7" reel | 13" reel |
| 0402 | 0.50±0.05 | N | 10K | 20K |
| 0603 | 0.80±0.07 | S | 4K | 10K |

Unit: pieces

## ELECTRICAL CHARACTERISTICS

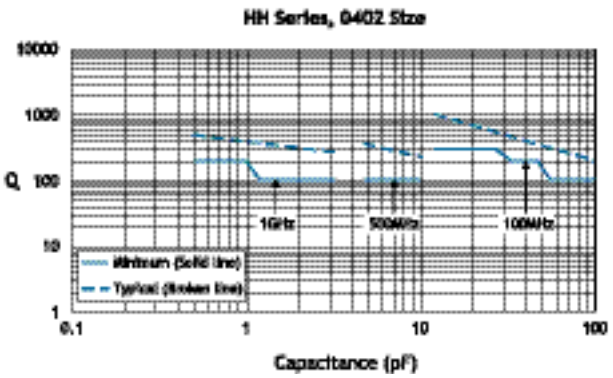



Fig. 2 Q factor specification vs. Specific frequency for 0402

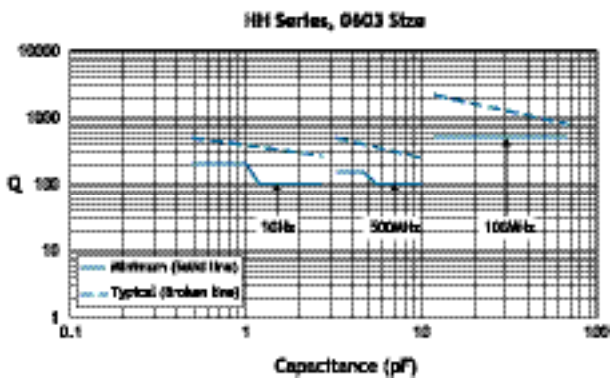



Fig. 3 Q factor specification vs. Specific frequency for 0603

# High Q and Low ESR Capacitors

## ■ CAPACITANCE RANGE

| Dielectric | | NP0 | | | | |
|---|---|---|---|---|---|---|
| Size | | 0402 | | 0603 | | |
| Rated Voltage (VDC) | | 16 | 50 | 16 | 50 | 100 |
| Capacitance | 0.5pF (0R5) | | N | | S | S |
| | 0.6pF (0R6) | | N | | S | S |
| | 0.7pF (0R7) | | N | | S | S |
| | 0.8pF (0R8) | | N | | S | S |
| | 0.9pF (0R9) | | N | | S | S |
| | 1.0pF (1R0) | | N | | S | S |
| | 1.2pF (1R2) | | N | | S | S |
| | 1.5pF (1R5) | | N | | S | S |
| | 1.8pF (1R8) | | N | | S | S |
| | 2.2pF (2R2) | | N | | S | S |
| | 2.7pF (2R7) | | N | | S | S |
| | 3.3pF (3R3) | | N | | S | S |
| | 3.9pF (3R9) | | N | | S | S |
| | 4.7pF (4R7) | | N | | S | S |
| | 5.6pF (5R6) | | N | | S | S |
| | 6.8pF (6R8) | | N | | S | S |
| | 8.2pF (8R2) | | N | | S | S |
| | 10pF (100) | | N | | S | S |
| | 12pF (120) | | N | | S | S |
| | 15pF (150) | | N | | S | S |
| | 18pF (180) | | N | | S | S |
| | 22pF (220) | | N | | S | S |
| | 27pF (270) | | N | | S | S |
| | 33pF (330) | | N | | S | S |
| | 39pF (390) | | N | | S | S |
| | 47pF (470) | | N | | S | S |
| | 56pF (560) | | N | | S | S |
| | 68pF (680) | | N | | S | S |
| | 82pF (820) | | N | | S | S |
| | 100pF (101) | | N | | S | S |
| | 120pF (121) | | N | | S | S |
| | 150pF (151) | | N | | S | S |
| | 180pF (181) | | N | | S | S |
| | 220pF (221) | | N | | S | S |
| | 270pF (271) | N | | | S | S |
| | 330pF (331) | N | | | S | S |
| | 390pF (391) | N | | | S | S |
| | 470pF (471) | N | | | S | S |
| | 560pF (561) | | | | S | |
| | 680pF (681) | | | | S | |
| | 820pF (821) | | | | S | |
| | 1,000pF (102) | | | | S | |
| | 1,200pF (122) | | | S | | |
| | 1,500pF (152) | | | S | | |
| | 1,800pF (182) | | | S | | |
| | 2,200pF (222) | | | S | | |
| | 2,700pF (272) | | | S | | |
| | 3,300pF (332) | | | S | | |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Microwave Capacitors

## HOW TO ORDER

| MW | 15 | N | 100 | G | 500 | L | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| MW=Microwave | 15=0402 (1005)<br><br>18=0603 (1608) | N=NP0 (C0J) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>R47=0.47pF<br>0R5=0.5pF<br>1R0=1.0pF<br>100=10x10⁰ =10pF | A=±0.05pF<br>B=±0.1pF<br>C=±0.25pF<br>F=±1%<br>G=±2%<br>J=±5%<br><br>(A for Cap<5pF; B,C,D for Cap<10pF; F,G,J for Cap≥10pF) | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>500=50 VDC | L=Ag/Ni/Sn | B=Bulk<br>C=Bulk cassette<br>T=7" reeled<br>G=13" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | |
|---|---|---|---|---|
| | | | 7" reel | 13" reel |
| 0402 | 0.50±0.05 | N | 10K | 20K |
| 0603 | 0.80±0.07 | S | 4K | 10K |

Unit: pieces

## ELECTRICAL CHARACTERISTICS

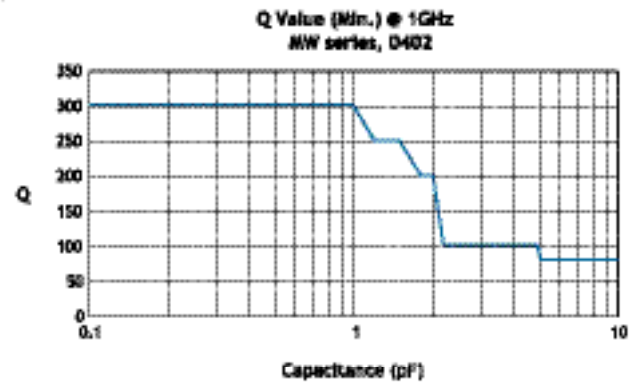



Fig. 2 Q value criteria vs. capacitance for 0402

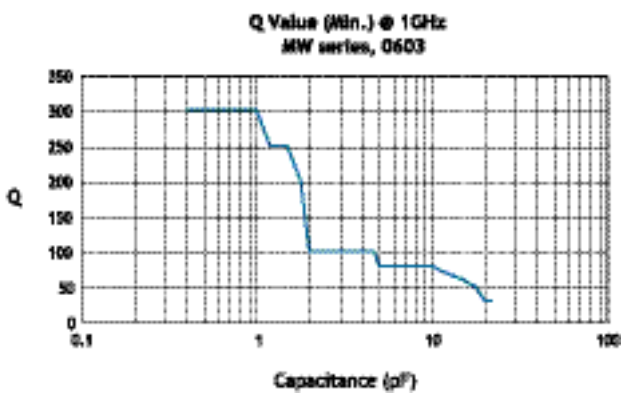



Fig. 3 Q value criteria vs. capacitance for 0603

# Microwave Capacitors

## ■ CAPACITANCE RANGE

| Dielectric | | NP0 | |
|---|---|---|---|
| Size | | 0402 | 0603 |
| Rated Voltage (VDC) | | 50 | 50 |
| Capacitance | 0.1pF (0R1) | N | |
| | 0.2pF (0R2) | N | |
| | 0.3pF (0R3) | N | |
| | 0.4pF (0R4) | N | S |
| | 0.5pF (0R5) | N | S |
| | 0.6pF (0R6) | N | S |
| | 0.7pF (0R7) | N | S |
| | 0.8pF (0R8) | N | S |
| | 0.9pF (0R9) | N | S |
| | 1.0pF (1R0) | N | S |
| | 1.2pF (1R2) | N | S |
| | 1.5pF (1R5) | N | S |
| | 1.8pF (1R8) | N | S |
| | 2.0pF (2R0) | N | S |
| | 2.2pF (2R2) | N | S |
| | 2.7pF (2R7) | N | S |
| | 3.0pF (3R0) | N | S |
| | 3.3pF (3R3) | N | S |
| | 3.9pF (3R9) | N | S |
| | 4.0pF (4R0) | N | S |
| | 4.7pF (4R7) | N | S |
| | 5.0pF (5R0) | N | S |
| | 5.6pF (5R6) | N | S |
| | 6.0pF (6R0) | N | S |
| | 6.8pF (6R8) | N | S |
| | 7.0pF (7R0) | N | S |
| | 8.0pF (8R0) | N | S |
| | 8.2pF (8R2) | N | S |
| | 9.0pF (9R0) | N | S |
| | 10pF (100) | N | S |
| | 12pF (120) | | S |
| | 15pF (150) | | S |
| | 18pF (180) | | S |
| | 22pF (220) | | S |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Open-mode Design Capacitors

## HOW TO ORDER

| OP | 32 | B | 103 | K | 201 | C | T |
|---|---|---|---|---|---|---|---|
| **Series** | **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| OP=Open-mode | 21=0805 (2012)<br><br>31=1206 (3216)<br><br>32=1210 (3225)<br><br>43=1812 (4532) | B=X7R | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>$102=10x10^2$<br>=1000pF | K=±10%<br>M=±20% | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>101=100 VDC<br>201=200 VDC<br>251=250 VDC<br>501=500 VDC | L=Ag/Ni/Sn<br>C=Cu/Ni/Sn<br>(Note 1) | B=Bulk<br>T=7" reeled<br>G=13" reeled |

Note 1: Please see below product range table to find right termination code.

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | | Plastic tape | |
|---|---|---|---|---|---|---|
| | | | 7" reel | 13" reel | 7" reel | 13" reel |
| 0805 | 0.80±0.10 | B | 4k | 15k | - | - |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| 1206 | 0.80±0.10 | B | 4k | 15k | - | - |
| | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| 1210 | 0.95±0.10 | C | - | - | 3k | 10k |
| | 1.25±0.10 | D | - | - | 3k | 10k |
| | 1.60±0.20 | G | - | - | 2k | - |
| | 2.50±0.30 | M | - | - | 1k | - |
| 1812 | 1.25±0.10 | D | - | - | 1k | - |
| | 2.00±0.20 | K | - | - | 1k | - |

Unit: pieces

## INNER CONSTRUCTION OF OPEN-MODE DESIGN

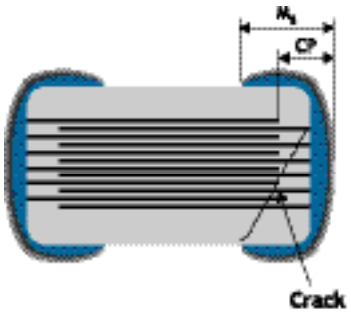



Fig. 2 Normal design (CP<$M_B$) – circuit leakage during cracking.

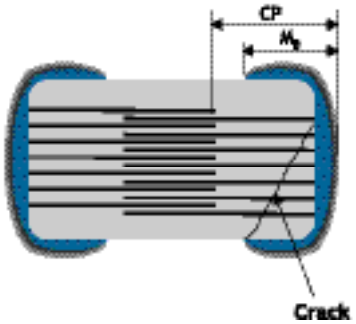



Fig. 3 Open-mode design (CP>$M_B$) – circuit open during cracking.

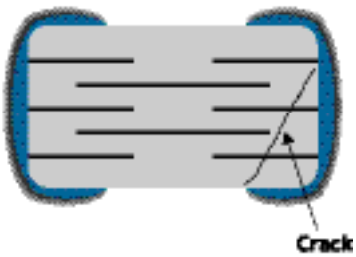



Fig. 4 Floating design (one kind of open-mode design)– circuit open during cracking.

# Open-mode Design Capacitors

## ■ CAPACITANCE RANGE

| Dielectric | X7R | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size | 0805 | | | | 1206 | | | | 1210 | | | | 1812 | | | |
| Rated Voltage (VDC) | 100 | 200 | 250 | 500 | 100 | 200 | 250 | 500 | 100 | 200 | 250 | 500 | 100 | 200 | 250 | 500 |
| Capacitance | | | | | | | | | | | | | | | | |
| 100pF (101) | B | B | B | B^ | | | | | | | | | | | | |
| 120pF (121) | B | B | B | B^ | | | | | | | | | | | | |
| 150pF (151) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 180pF (181) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 220pF (221) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 270pF (271) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 330pF (331) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 390pF (391) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 470pF (471) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 560pF (561) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 680pF (681) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 820pF (821) | B | B | B | B^ | B | B | B | B^ | | | | | | | | |
| 1,000pF (102) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 1,200pF (122) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 1,500pF (152) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 1,800pF (182) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 2,200pF (222) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 2,700pF (272) | B | B | B | B^ | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 3,300pF (332) | B | B | B | | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 3,900pF (392) | B | B | B | | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 4,700pF (472) | B | B | B | | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 5,600pF (562) | B | D | D | | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 6,800pF (682) | B | D | D | | B | B | B | B^ | C | C | C | C^ | D | D | D | D^ |
| 8,200pF (822) | B | D | D | | B | B | B | C^ | C | C | C | C^ | D | D | D | D^ |
| 0.010μF (103) | B | D | D | | B | B | B | C^ | C | C | C | C^ | D | D | D | D^ |
| 0.012μF (123) | B | D | D | | B | B | B | D^ | C | C | C | C^ | D | D | D | D^ |
| 0.015μF (153) | B | D | D | | B | C | C | D^ | C | C | C | C^ | D | D | D | D^ |
| 0.018μF (183) | B | D | D | | B | C | C | D^ | C | C | C | C^ | D | D | D | D^ |
| 0.022μF (223) | B | D | D | | B | C | C | G^ | C | C | C | D^ | D | D | D | D^ |
| 0.027μF (273) | D | | | | B | C | C | G^ | C | C | C | G^ | D | D | D | D^ |
| 0.033μF (333) | D | | | | B | G | G | G^ | C | C | C | G^ | D | D | D | D^ |
| 0.039μF (393) | D | | | | B | G | G | | C | C | C | G^ | D | D | D | D^ |
| 0.047μF (473) | D | | | | B | G | G | | C | D | D | G^ | D | D | D | D^ |
| 0.056μF (563) | | | | | B | G | G | | C | D | D | G^ | D | D | D | K^ |
| 0.068μF (683) | | | | | B | G | G | | C | G | G | | D | D | D | K^ |
| 0.082μF (823) | | | | | B | G | G | | C | G | G | | D | D | D | K^ |
| 0.10μF (104) | | | | | D | G | G | | C | G | G | | D | D | D | K^ |
| 0.12μF (124) | | | | | D | | | | C | G | G | | D | D | D | |
| 0.15μF (154) | | | | | G | | | | D | M | M | | D | K | K | |
| 0.18μF (184) | | | | | G | | | | D | M | M | | D | K | K | |
| 0.22μF (224) | | | | | G | | | | D | M | M | | D | K | K | |
| 0.27μF (274) | | | | | | | | | G | | | | D | K | K | |
| 0.33μF (334) | | | | | | | | | G | | | | D | K | K | |
| 0.39μF (394) | | | | | | | | | M | | | | D | K | K | |
| 0.47μF (474) | | | | | | | | | M | | | | K | K | K | |
| 0.56μF (564) | | | | | | | | | M | | | | K | | | |
| 0.68μF (684) | | | | | | | | | | | | | K | | | |
| 0.82μF (824) | | | | | | | | | | | | | K | | | |
| 1.0μF (105) | | | | | | | | | | | | | K | | | |

1. The letter in cell is expressed the symbol of product thickness.
2. The letter in cell with "^" mark is expressed product with Ag/Ni/Sn terminations.
3. For more information about products with special capacitance or other data, please contact WTC local representative.

# Capacitor Arrays

## HOW TO ORDER

| Y | 4C | 3 | B | 103 | K | 500 | C | T |
|---|---|---|---|---|---|---|---|---|
| **Series** | **Cap. Nr.** | **Termination pitch** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Y=Capacitor Array | 4C=4xCap | 3=0.03” pitch | N=NP0 (C0G)<br>B=X7R<br>F=Y5V | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>103=10x10³<br>=10,000pF<br>=10nF | J=±5%<br>K=±10%<br>M=±20%<br>Z=+20/-80% | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>160=16 VDC<br>250=25 VDC<br>500=50 VDC | L=Ag/Ni/Sn (for NP0 dielectric)<br><br>C=Cu/Ni/Sn (for X7R, Y5V dielectric) | T=7” reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | Paper tape | |
|---|---|---|---|---|
| | | | 7” reel | 13” reel |
| 4 x 0603 | 0.80±0.10 | B | 4k | - |

Unit: pieces

## CAPACITANCE RANGE

| Size | | 4 x 0603 | | | |
|---|---|---|---|---|---|
| Dielectric | | NP0 | X7R | | Y5V |
| Rated Voltage (VDC) | | 50 | 16 | 50 | 50 |
| Capacitance | 10pF (100) | B | | | |
| | 15pF (150) | B | | | |
| | 22pF (220) | B | | | |
| | 33pF (330) | B | | | |
| | 47pF (470) | B | | | |
| | 68pF (680) | B | | | |
| | 100pF (101) | B | | | |
| | 150pF (151) | B | | | |
| | 180pF (181) | B | | B | |
| | 220pF (221) | B | | B | |
| | 330pF (331) | B | | B | |
| | 470pF (471) | B | | B | |
| | 1,000pF (102) | | | B | |
| | 1,500pF (152) | | | B | |
| | 2,200pF (222) | | | B | |
| | 3,300pF (332) | | | B | |
| | 4,700pF (472) | | | B | |
| | 6,800pF (682) | | | B | |
| | 0.010μF (103) | | | B | B |
| | 0.015μF (153) | | B | B | |
| | 0.022μF (223) | | B | B | B |
| | 0.033μF (333) | | B | | |
| | 0.047μF (473) | | B | | B |
| | 0.068μF (683) | | B | | |
| | 0.10μF (104) | | B | | B |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Low Inductance Capacitors

## HOW TO ORDER

| 0612 | B | 103 | K | 500 | C | T |
|---|---|---|---|---|---|---|
| **Size** | **Dielectric** | **Capacitance** | **Tolerance** | **Rated voltage** | **Termination** | **Packaging** |
| Inch (mm)<br>0612 (1632) | B=X7R | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>$103=10x10^3$<br>=10nF | K=±10%<br>M=±20% | Two significant digits followed by no. of zeros. And R is in place of decimal point.<br><br>eg.:<br>500=50VDC | C=Cu/Ni/Sn | T=7" reeled |

## PACKAGING DIMENSION AND QUANTITY

| Size | Thickness (mm)/Symbol | | 7" reel / Paper tape |
|---|---|---|---|
| 0612 (1632) | 0.80±0.10 | B | 4k |

Unit: pieces

## CAPACITANCE RANGE

| Dielectric | | X7R |
|---|---|---|
| Size | | 0612 |
| Rated Voltage (VDC) | | 50 |
| Capacitance | 10nF (103) | B |
| | 12nF (123) | B |
| | 15nF (153) | B |
| | 18nF (183) | B |
| | 22nF (223) | B |
| | 27nF (273) | B |
| | 33nF (333) | B |
| | 39nF (393) | B |
| | 47nF (473) | B |
| | 56nF (563) | B |
| | 68nF (683) | B |
| | 82nF (823) | B |
| | 100nF (104) | B |
| | 120nF (124) | B |
| | 150nF (154) | B |

1. The letter in cell is expressed the symbol of product thickness.
2. For more information about products with special capacitance or other data, please contact WTC local representative.

# Appendix I : Reliability Test Conditions and Requirements

<table>
<tr><th>NO.</th><th>Item</th><th>Test Condition</th><th>Requirements</th></tr>
<tr><td>1.</td><td>Visual and Mechanical</td><td>---</td><td>* No remarkable defect.<br>* Dimensions to confirm to individual specification sheet.</td></tr>
<tr><td>2.</td><td>Capacitance</td><td rowspan="2">Class I : NP0<br>Cap≤1000pF 1.0±0.2Vrms, 1MHz±10%<br>Cap>1000pF 1.0±0.2Vrms, 1KHz±10%<br><br>Class II : X7R, X5R, Y5V<br>Cap≤10μF, 1.0±0.2Vrms, 1KHz±10%<br>Cap>10μF, 0.5±0.2Vrms, 120Hz±20%</td><td>* Shall not exceed the limits given in the detailed spec.</td></tr>
<tr><td>3.</td><td>Q/ D.F.<br>(Dissipation Factor)</td><td>NP0: Cap≥30pF, Q≥1000; Cap<30pF, Q≥400+20C<br>X7R, X5R:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>2.5%</td><td>3.0%</td><td>0201(50V);0603≥0.047μF;0805≥0.18μF;<br>1206≥0.47μF</td></tr>
<tr><td rowspan="3">25V</td><td rowspan="3">3.5%</td><td>5.0%</td><td>0805≥1μF, 1210≥10μF</td></tr>
<tr><td>7.0%</td><td>0603≥0.33μF;TT series & Cap≥1μF</td></tr>
<tr><td>10%</td><td>1206≥10μF</td></tr>
<tr><td rowspan="2">16V</td><td rowspan="2">3.5%</td><td>5.0%</td><td>0402≥0.033μF;0603≥0.15μF;<br>0805≥0.68μF;1206≥2.2μF;1210≥4.7μF</td></tr>
<tr><td>10%</td><td>TT series & Cap≥1μF;1210≥22μF</td></tr>
<tr><td>10V</td><td>5.0%</td><td>10.0%</td><td>TT series & Cap≥1μF;0805≥2.2μF;<br>0603≥1μF;1206≥10μF;1210≥100μF</td></tr>
<tr><td>6.3V</td><td>10.0%</td><td>15.0%</td><td>0805≥10μF;1210≥100μF;<br>TT series & Cap≥1μF</td></tr>
</table>
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>5.0%</td><td>---</td><td>---</td></tr>
<tr><td>35V</td><td>7.0%</td><td>---</td><td>---</td></tr>
<tr><td rowspan="2">25V</td><td rowspan="2">5.0%</td><td>7.0%</td><td>0603≥0.1μF; 0805≥0.33μF;<br>1206≥1μF; 1210≥4.7μF</td></tr>
<tr><td>9.0%</td><td>0402≥0.068μF</td></tr>
<tr><td>16V (C<1.0μF)</td><td>7.0%</td><td>9.0%</td><td>0402≥0.068μF; 0603≥0.68μF</td></tr>
<tr><td>16V (C≥1.0μF)</td><td>9.0%</td><td>12.5%</td><td>0805≥4.7μF;1206≥10μF;1210≥22μF;<br>1812≥47μF;TT series & Cap≥1μF</td></tr>
<tr><td>10V</td><td>12.5%</td><td>---</td><td>---</td></tr>
<tr><td>6.3V</td><td>20.0%</td><td>---</td><td>---</td></tr>
</table></td></tr>
<tr><td>4a.</td><td>Dielectric Strength</td><td>* To apply voltage (≤50V) 250%.<br>* Duration : 1 to 5 sec.<br>* Charge & discharge current less than 50mA.</td><td rowspan="2">* No evidence of damage or flash over during test.</td></tr>
<tr><td></td><td></td><td>* To apply voltage :<br>100V ≥3 times V DC<br>200V ~ 300V ≥2 times V DC<br>500V ~ 999V ≥1.5 times V DC<br>1000V ~ 3000V ≥1.2 times V DC<br>* Cut-off, set at 10mA<br>* TEST= 15 sec.<br>* RAMP=0</td></tr>
<tr><td>4b.</td><td>Dielectric Strength<br>(for X1/Y2 & X2/Y3)</td><td>* To apply 1500 VAC voltage.<br>* Duration: 60 sec.</td><td>* No evidence of damage or flash over during test.</td></tr>
<tr><td rowspan="3">5.</td><td rowspan="3">Insulation Resistance</td><td colspan="2">To apply rated voltage for max. 120 sec.</td><td>≥10GΩ or RxC≥500Ω-F whichever is smaller.<br>6.3V: RxC≥100Ω-F</td></tr>
<tr><td>Rated voltage:<br>200 ~ 630V</td><td>To apply rated voltage (500V max)<br>for 60 sec</td><td>≥10GΩ or RxC≥100Ω-F whichever is smaller.</td></tr>
<tr><td>Rated voltage:<br>> 630V</td><td>To apply 500V for 60 sec.</td><td>≥10GΩ</td></tr>
<tr><td>6.</td><td>Temperature Coefficient</td><td>With no electrical load.
<table>
<tr><th>T.C.</th><th>Operating Temp</th></tr>
<tr><td>NP0 (C0G)</td><td>-55~125°C at 25°C</td></tr>
<tr><td>NP0 (C0J)</td><td>-55~125°C at 25°C</td></tr>
<tr><td>X7R</td><td>-55~125°C at 25°C</td></tr>
<tr><td>X5R</td><td>-55~85°C at 25°C</td></tr>
<tr><td>Y5V</td><td>-25~85°C at 20°C</td></tr>
</table></td><td>
<table>
<tr><th>T.C.</th><th>Capacitance Change</th></tr>
<tr><td>NP0 (C0G)</td><td>Within ±30ppm/°C</td></tr>
<tr><td>NP0 (C0J)</td><td>Within ±120ppm/°C</td></tr>
<tr><td>X7R</td><td>Within ±15%</td></tr>
<tr><td>X5R</td><td>Within ±15%</td></tr>
<tr><td>Y5V</td><td>Within +30%/-80%</td></tr>
</table></td></tr>
<tr><td>7.</td><td>Adhesive Strength of Termination</td><td>* Pressurizing force:<br>0201: 2N<br>0402 & 0603: 5N<br>>0603: 10N<br>* Test time: 10±1 sec.</td><td>* No remarkable damage or removal of the terminations.</td></tr>
<tr><td>8.</td><td>Vibration Resistance</td><td>* Vibration frequency: 10~55 Hz/min.<br>* Total amplitude: 1.5mm<br>* Test time: 6 hrs. (Two hrs each in three mutually perpendicular directions.)</td><td>* No remarkable damage.<br>* Cap change and Q/D.F.: To meet initial spec.</td></tr>
</table>

# Appendix I : Reliability Test Conditions and Requirements

<table>
<tr><th>NO.</th><th>Item</th><th>Test Condition</th><th>Requirements</th></tr>
<tr><td>9.</td><td>Solderability</td><td>* Solder temperature: 235±5°C<br>* Dipping time: 2±0.5 sec.</td><td>95% min. coverage of all metalized area.</td></tr>
<tr><td>10.</td><td>Bending Test</td><td>* The middle part of substrate shall be pressurized by means of the pressurizing rod at a rate of about 1 mm per second until the deflection becomes 1 mm and then the pressure shall be maintained for 5±1 sec.<br>* Measurement to be made after keeping at room temp. for 24±2 hrs.</td><td>* No remarkable damage.<br>* Cap change:<br>NP0: within ±5.0% or ±0.5pF whichever is larger.<br>X7R, X5R: within ±12.5%<br>Y5V: within ±30%<br>(This capacitance change means the change of capacitance under specified flexure of substrate from the capacitance measured before the test.)</td></tr>
<tr><td>11.</td><td>Resistance to Soldering Heat</td><td>* Solder temperature: 270±5°C<br>* Dipping time: 10±1 sec<br>* Preheating: 120 to 150°C for 1 minute before immerse the capacitor in an eutectic solder.<br>* Before initial measurement (Class II only): Perform 150+0/-10°C for 1 hr and then set for 48±4 hrs at room temp.<br>* Measurement to be made after keeping at room temp. for 24±2 hrs. (Class I) or 48±4 hrs. (Class II).</td><td>* No remarkable damage.<br>* Cap change:<br>NP0: within ±2.5% or ±0.25pF whichever is larger.<br>X7R, X5R: within ±7.5%<br>Y5V: within ±20%<br>* Q/D.F., I.R. and dielectric strength: To meet initial requirements.<br>* 25% max. leaching on each edge.</td></tr>
<tr><td>12.</td><td>Temperature Cycle</td><td>* Conduct the five cycles according to the temperatures and time.
<table>
<tr><th>Step</th><th>Temp. (°C)</th><th>Time (min.)</th></tr>
<tr><td>1</td><td>Min. operating temp. +0/-3</td><td>30±3</td></tr>
<tr><td>2</td><td>Room temp.</td><td>2~3</td></tr>
<tr><td>3</td><td>Max. operating temp. +3/-0</td><td>30±3</td></tr>
<tr><td>4</td><td>Room temp.</td><td>2~3</td></tr>
</table>
* Before initial measurement (Class II only): Perform 150+0/-10°C for 1 hr and then set for 48±4 hrs at room temp.<br>* Measurement to be made after keeping at room temp. for 24±2 hrs. (Class I) or 48±4 hrs. (Class II).</td><td>* No remarkable damage.<br>* Cap change:<br>NP0: within ±2.5% or ±0.25pF whichever is larger.<br>X7R, X5R: within ±7.5%<br>Y5V: within ±20%<br>* Q/D.F., I.R. and dielectric strength: To meet initial requirements.</td></tr>
<tr><td>13.</td><td>Humidity (Steady State)</td><td>* Test temp.: 40±2°C<br>* Humidity: 90~95% RH<br>* Test time: 500+24/-0hrs.<br>* Measurement to be made after keeping at room temp. for 24±2 hrs. (Class I) or 48±4 hrs. (Class II).</td><td>* No remarkable damage.<br>* Cap change: NP0: within ±5.0% or ±0.5pF whichever is larger.<br>X7R, X5R: ≥10V, within ±12.5%, 6.3V, within ±25%<br>TT series & Cap≥1μF, within ±25%<br>Y5V: ≥10V, within ±30%,<br>6.3V, within ±25%<br>* Q/D.F. value:<br>NP0: Cap≥30pF, Q≥350; 10pF≤Cap<30pF, Q≥275+2.5C<br>Cap<10pF; Q≥200+10C<br>X7R, X5R:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>3.0%</td><td>6.0%</td><td>0603≥0.047μF; 0805≥0.18μF; 0201(50V) 1206≥0.47μF</td></tr>
<tr><td>25V</td><td>5.0%</td><td>10.0%</td><td>0805≥1μF, 1210≥10μF</td></tr>
<tr><td></td><td></td><td>14.0%</td><td>0603≥0.33μF; TT series & Cap≥1μF</td></tr>
<tr><td>16V</td><td>5.0%</td><td>10.0%</td><td>0402≥0.033μF; 0603≥0.15μF; 0805≥0.68μF; 1206≥2.2μF; TT series & Cap≥1μF</td></tr>
<tr><td>10V</td><td>7.5%</td><td>15.0%</td><td>0402≥0.056μF; 0603≥0.33μF; 0805≥2.2μF; 1206≥2.2μF; 1210≥22μF TT series & Cap≥1μF</td></tr>
<tr><td>6.3V</td><td>15.0%</td><td>30.0%</td><td>0805≥10μF; 1210≥100μF; TT series & Cap≥1μF</td></tr>
</table>
Y5V:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>7.5%</td><td>---</td><td>---</td></tr>
<tr><td>35V</td><td>10.0%</td><td>---</td><td>---</td></tr>
<tr><td>25V</td><td>7.5%</td><td>10.0%</td><td>0603≥0.1μF; 0805≥0.33μF; 1206≥1μF; 1210≥4.7μF</td></tr>
<tr><td></td><td></td><td>12.5%</td><td>0402≥0.068μF</td></tr>
<tr><td>16V (C<1.0μF)</td><td>10.0%</td><td>12.5%</td><td>0402≥0.068μF; 0603≥0.68μF</td></tr>
<tr><td>16V (C≥1.0μF)</td><td>12.5%</td><td>20.0%</td><td>0805≥4.7μF;1206≥10μF; 1210≥22μF; 1812≥47μF</td></tr>
<tr><td>10V</td><td>15.0%</td><td>---</td><td>---</td></tr>
<tr><td>6.3V</td><td>30.0%</td><td>---</td><td>---</td></tr>
</table>
* I.R.: ≥10V 1GΩ or 50Ω-F whichever is smaller.<br>6.3V, 10Ω-F</td></tr>
</table>

# Appendix I : Reliability Test Conditions and Requirements

<table>
<tr><th>NO.</th><th>Item</th><th>Test Condition</th><th>Requirements</th></tr>
<tr><td>14.</td><td>Humidity Load<br>(Damp Heat)</td><td>* Test temp.: 40±2°C<br>* Humidity: 90~95%RH<br>* Test time: 500+24/-0 hrs.<br>* To apply voltage: rated voltage (Max. 500V)<br>* Measurement to be made after keeping at room temp. for 24±2 hrs. (Class I) or 48±4 hrs. (Class II).</td><td>* No remarkable damage.<br>* Cap change: NP0: within ±7.5% or ±0.75pF whichever is larger.<br>X7R, X5R: ≥10V, within ±12.5%, 6.3V, within ±25%<br>TT series & Cap≥1μF, within ±25%<br>Y5V: ≥10V, within ±30%<br>6.3V, within +30 to -40%<br>* Q/D.F. value:<br>NP0: Cap≥30pF, Q≥200; Cap<30pF, Q≥100+10/3C<br>X7R, X5R:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>3.0%</td><td>6.0%</td><td>0603≥0.047μF;0805≥0.18μF;1206≥0.47μF<br>0201(50V)</td></tr>
<tr><td rowspan="2">25V</td><td rowspan="2">5.0%</td><td>10.0%</td><td>0805≥1μF, 1210≥10μF</td></tr>
<tr><td>14.0%</td><td>0603≥0.33μF; TT series & Cap≥1μF</td></tr>
<tr><td>16V</td><td>5.0%</td><td>10.0%</td><td>0402≥0.033μF; 0603≥0.15μF;<br>0805≥0.68μF; 1206≥2.2μF; TT series & Cap≥1μF</td></tr>
<tr><td>10V</td><td>7.5%</td><td>15.0%</td><td>0402≥0.056μF; 0603≥0.33μF;<br>0805≥2.2μF; 1206≥2.2μF;1210≥22μF<br>TT series & Cap≥1μF</td></tr>
<tr><td>6.3V</td><td>15.0%</td><td>30.0%</td><td>0805≥10μF;1210≥100μF: TT Serles & Cap≥1μF</td></tr>
</table>
Y5V:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
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<tr><td>12.5%</td><td>0402≥0.068μF</td></tr>
<tr><td>16V (C<1.0μF)</td><td>10.0%</td><td>12.5%</td><td>0402≥0.068μF; 0603≥0.68μF</td></tr>
<tr><td>16V (C≥1.0μF)</td><td>12.5%</td><td>20.0%</td><td>0805≥4.7μF; 1206≥10μF;1210≥22μF;<br>1812≥47μF</td></tr>
<tr><td>10V</td><td>15.0%</td><td>---</td><td>---</td></tr>
<tr><td>6.3V</td><td>30.0%</td><td>---</td><td>---</td></tr>
</table>
* I.R.: ≥10V,500MΩ or 25Ω-F whichever is smaller.<br>6.3V, 5Ω-F</td></tr>
<tr><td>15.</td><td>High Temperature Load (Endurance)</td><td>* Test temp.:<br>NP0, X7R: 125±3°C<br>X5R, Y5V: 85±3°C<br>* To apply voltage:<br>(1) 6.3V or C≥10μF (for X7R, X5R): 150% of rated voltage.<br>(2) 6.3V<V<500V and C<10μF (for X7R, X5R): 200% of rated voltage.<br>(3) 500V: 150% of rated voltage.<br>(4) V≥630V: 120% of rated voltage.<br>* Test time: 1000+24/-0 hrs.<br>* Measurement to be made after keeping at room temp. for 24±2 hrs. (Class I) or 48±4 hrs. (Class II).</td><td>* No remarkable damage.<br>* Cap change: NP0: within ±3.0% or ±0.3pF whichever is larger.<br>X7R, X5R: ≥10V, within ±12.5%, 6.3V, within ±25%<br>TT series & Cap≥1μF, within ±25%<br>Y5V: ≥10V, within ±30%<br>6.3V, within +30 to -40%<br>* Q/D.F. value:<br>NP0: Cap≥30pF, Q≥350<br>10pF≤Cap<30pF, Q≥275+2.5C<br>Cap<10pF, Q≥200+10C<br>X7R, X5R:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>3.0%</td><td>6.0%</td><td>0603≥0.047μF; 0805≥0.18μF,<br>1206≥0.47μF; 0201(50V)</td></tr>
<tr><td rowspan="2">25V</td><td rowspan="2">5.0%</td><td>10.0%</td><td>0805≥1μF, 1210≥10μF</td></tr>
<tr><td>14.0%</td><td>0603≥0.33μF; TT series & Cap≥1μF</td></tr>
<tr><td>16V</td><td>5.0%</td><td>10.0%</td><td>0402≥0.033μF; 0603≥0.15μF;<br>0805≥0.68μF; 1206≥2.2μF; TT series & Cap≥1μF</td></tr>
<tr><td>10V</td><td>7.5%</td><td>15.0%</td><td>0402≥0.056μF; 0603≥0.33μF;<br>0805≥2.2μF; 1206≥2.2μF; 1210≥22μF<br>TT series & Cap≥1μF</td></tr>
<tr><td>6.3V</td><td>15.0%</td><td>30.0%</td><td>0805≥10μF; 1210≥100μF; TT series & Cap≥1μF</td></tr>
</table>
Y5V:
<table>
<tr><th>Rated vol.</th><th>D.F.≤</th><th colspan="2">Exception of D.F.≤</th></tr>
<tr><td>≥50V</td><td>7.5%</td><td>---</td><td>---</td></tr>
<tr><td>35V</td><td>10.0%</td><td>---</td><td>---</td></tr>
<tr><td rowspan="2">25V</td><td rowspan="2">7.5%</td><td>10.0%</td><td>0603≥0.1μF; 0805≥0.33μF;<br>1206≥1μF; 1210≥4.7μF</td></tr>
<tr><td>12.5%</td><td>0402≥0.068μF</td></tr>
<tr><td>16V (C<1.0μF)</td><td>10.0%</td><td>12.5%</td><td>0402≥0.068μF; 0603≥0.68μF</td></tr>
<tr><td>16V (C≥1.0μF)</td><td>12.5%</td><td>20.0%</td><td>0805≥4.7μF; 1206≥10μF;1210≥22μF;<br>1812≥47μF</td></tr>
<tr><td>10V</td><td>15.0%</td><td>---</td><td>---</td></tr>
<tr><td>6.3V</td><td>30.0%</td><td>---</td><td>---</td></tr>
</table>
* I.R.: ≥10V,1GΩ or 50Ω-F whichever is smaller.<br>6.3V, 10Ω-F</td></tr>
</table>

# Appendix II : General Information

## ■ Constructions

| No. | Name | | NP0/X7R | X7R/X5R/Y5V |
|---|---|---|---|---|
| ① | Ceramic material | | $BaTiO_3$ based | |
| ② | Inner electrode | | AgPd alloy | Ni |
| ③ | Termination | Inner layer | Ag | Cu |
| ④ | | Middle layer | Ni | |
| ⑤ | | Outer layer | Sn (Matt) | |

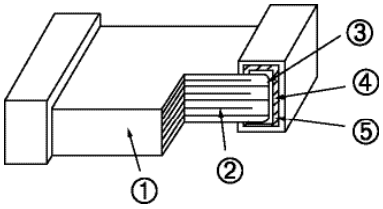


Fig.1 The construction of MLCC

## ■ Storage and handling conditions

(1) To store products at 5 to 40°C ambient temperature and 20 to 70%. related humidity conditions.

(2) The product is recommended to be used within one year after shipment. Check solderability in case of shelf life extension is needed.

Cautions:

a. Don't store products in a corrosive environment such as sulfide, chloride gas, or acid. It may cause oxidization of electrode, which easily be resulted in poor soldering.

b. To store products on the shelf and avoid exposure to moisture.

c. Don't expose products to excessive shock, vibration, direct sunlight and so on.

## ■ Recommended soldering conditions

The lead-free termination MLCCs are not only to be used on SMT against lead-free solder paste, but also suitable against lead-containing solder paste. If the optimized solder joint is requested, increasing soldering time, temperature and concentration of $N_2$ within oven are recommended.

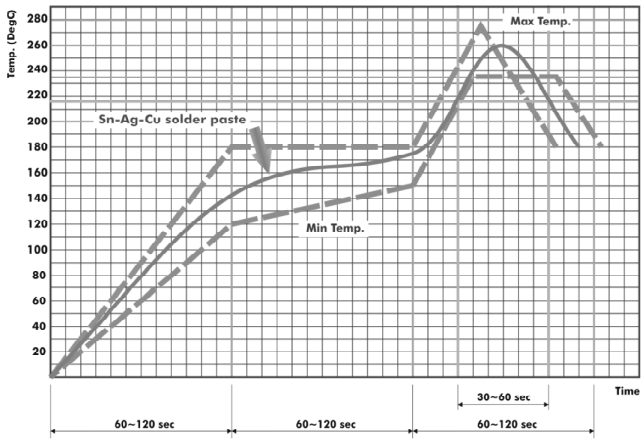


Fig. 2 IR reflow soldering profile for SMT process with SnAgCu series solder paste.

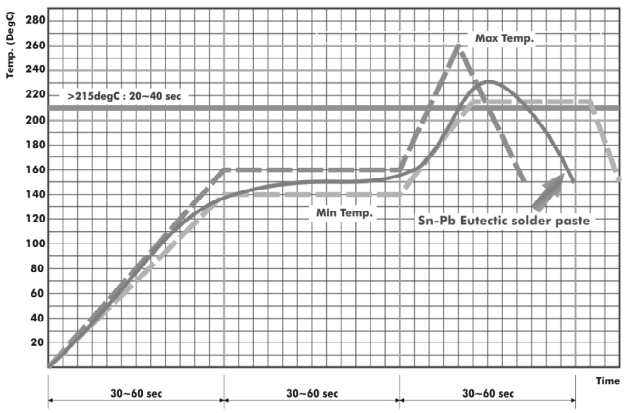


Fig. 3 IR reflow soldering profile for SMT process with eutectic SnPb solder paste.

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